

GreenDelta

sustainability consulting + software



openLCA System Dynamics Training

Alexander Koch

GreenDelta GmbH

Picture: Julia Cilleruego Palomero



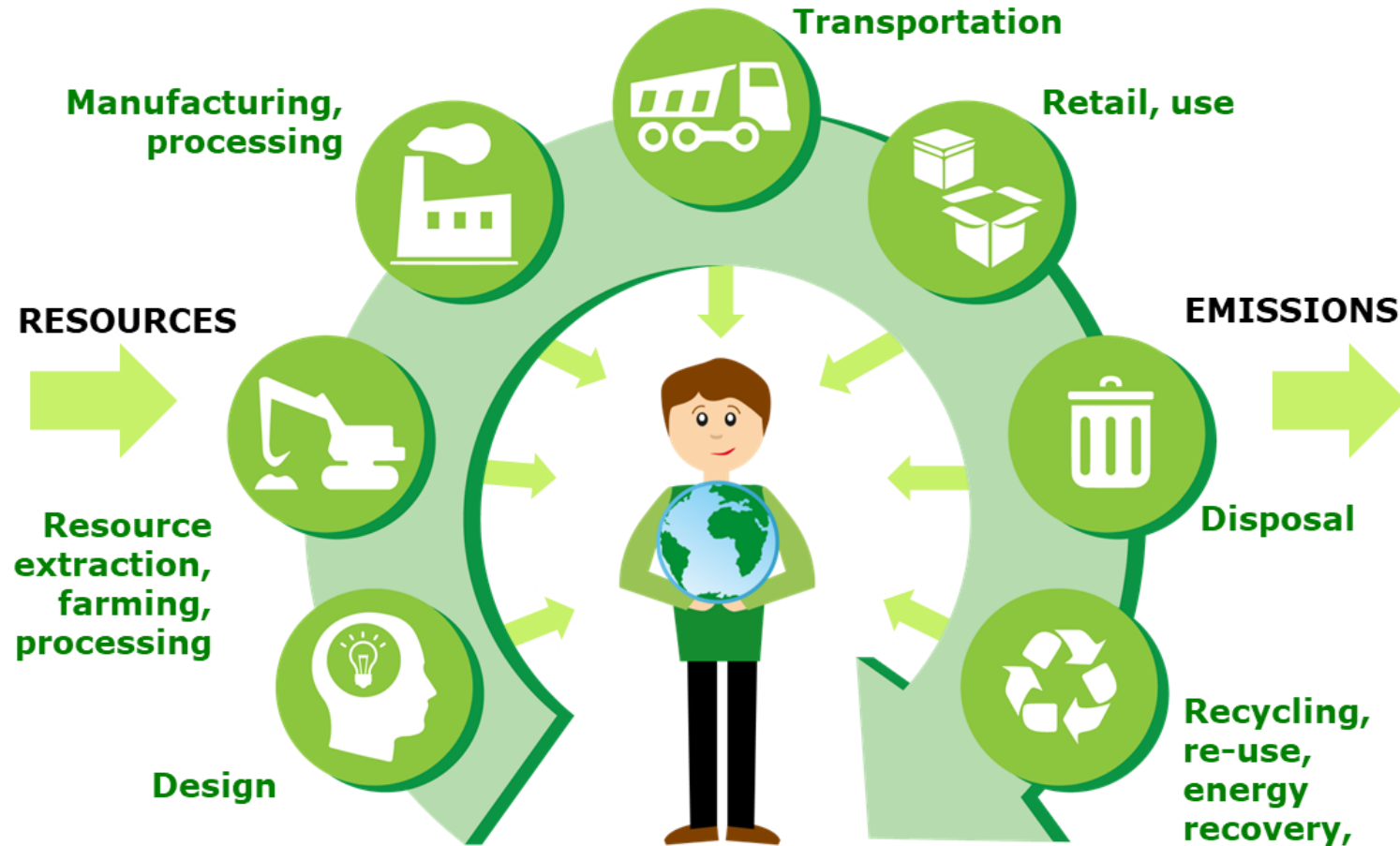
System dynamics training

- LCA today
- System dynamics (SD) theory
- Create and running SD models in openLCA
- Linking SD and LCA product systems
- Running simulations with impact results
- Export results and comparing scenarios

LCA today



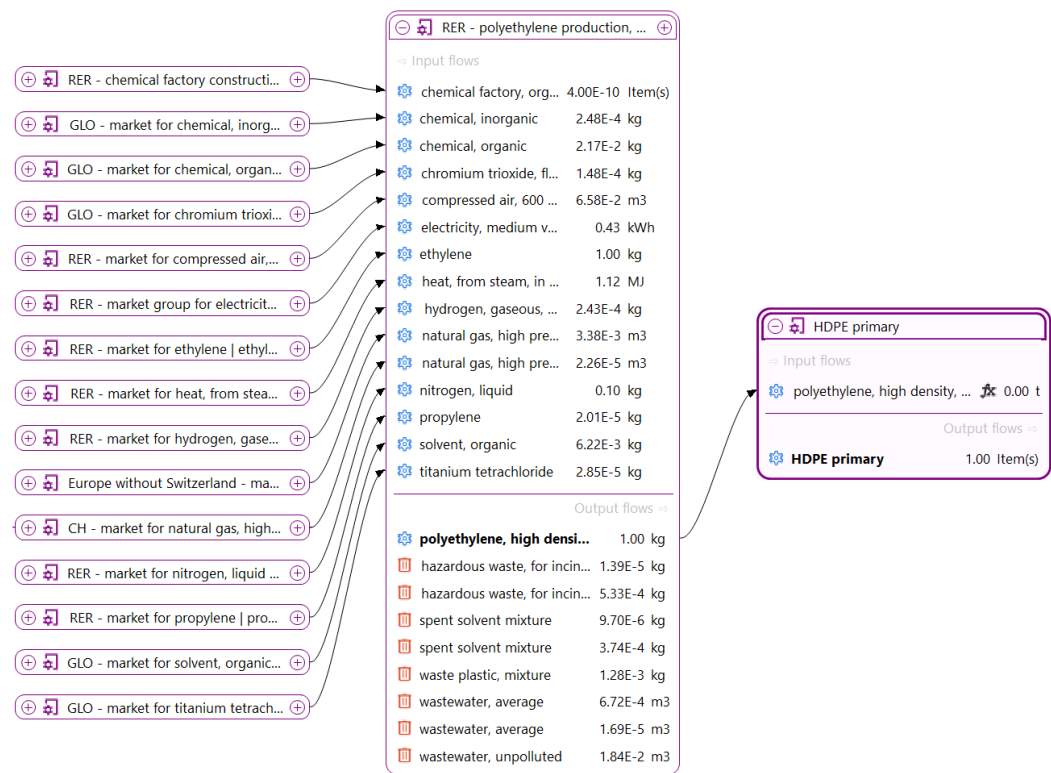
LCA today



Life Cycle Thinking

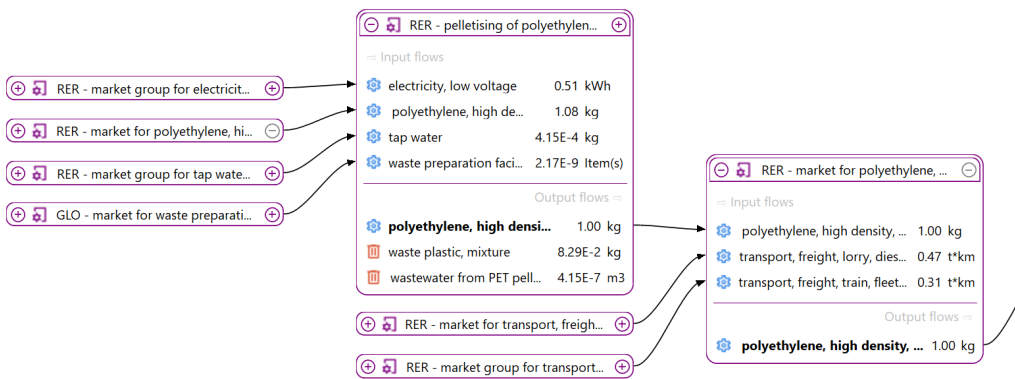
LCA today

Virgin plastics

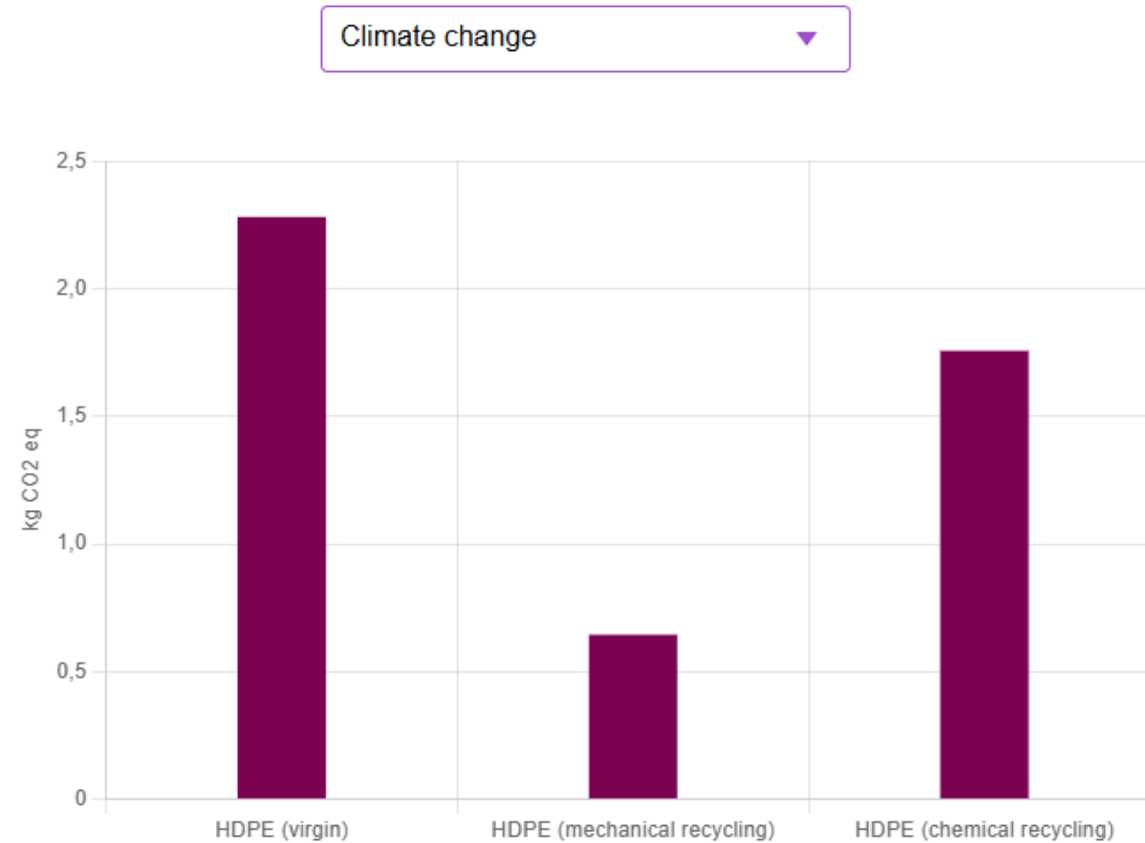


vs.

Recycled plastics

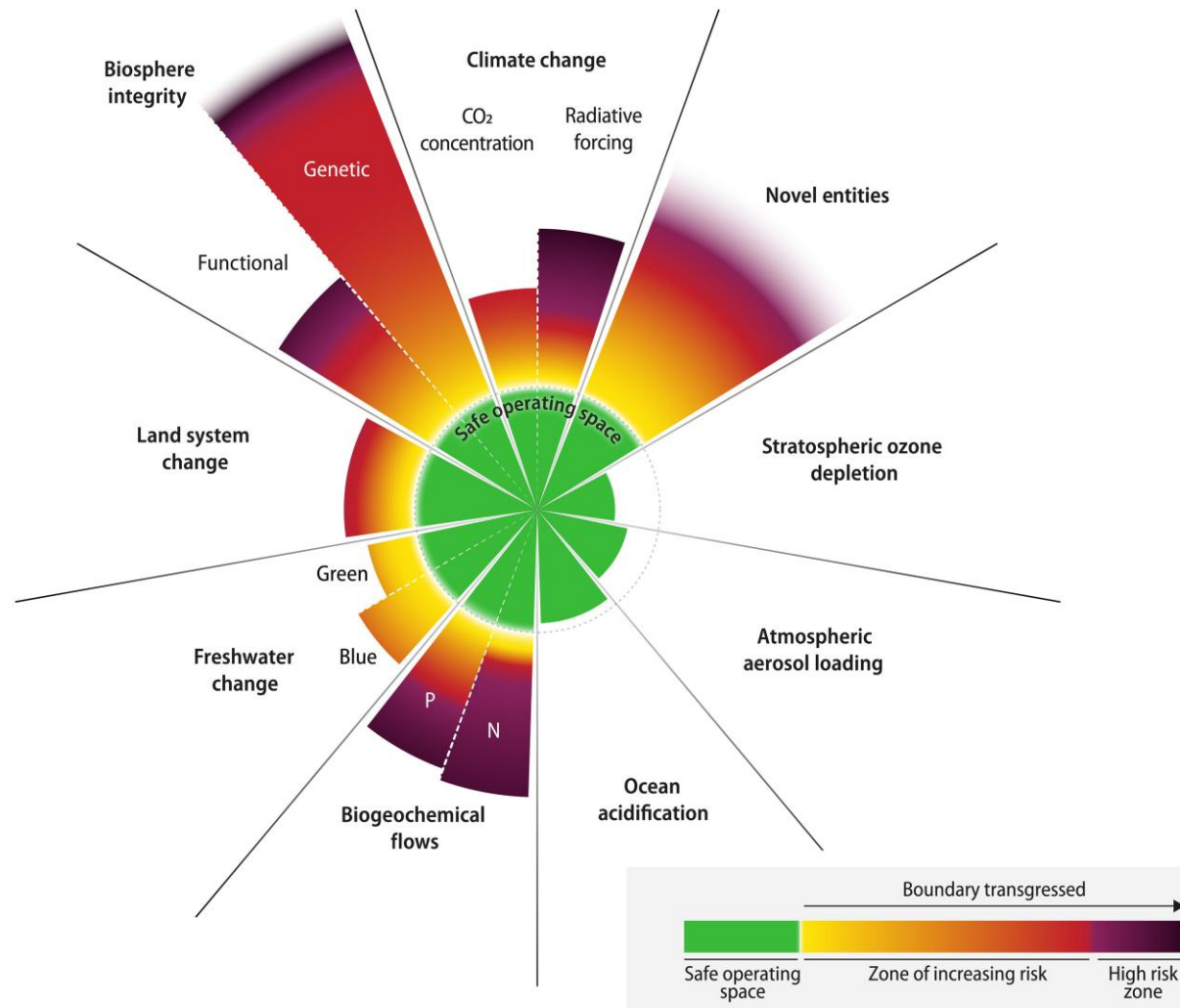


Comparing alternatives for establishing preferences



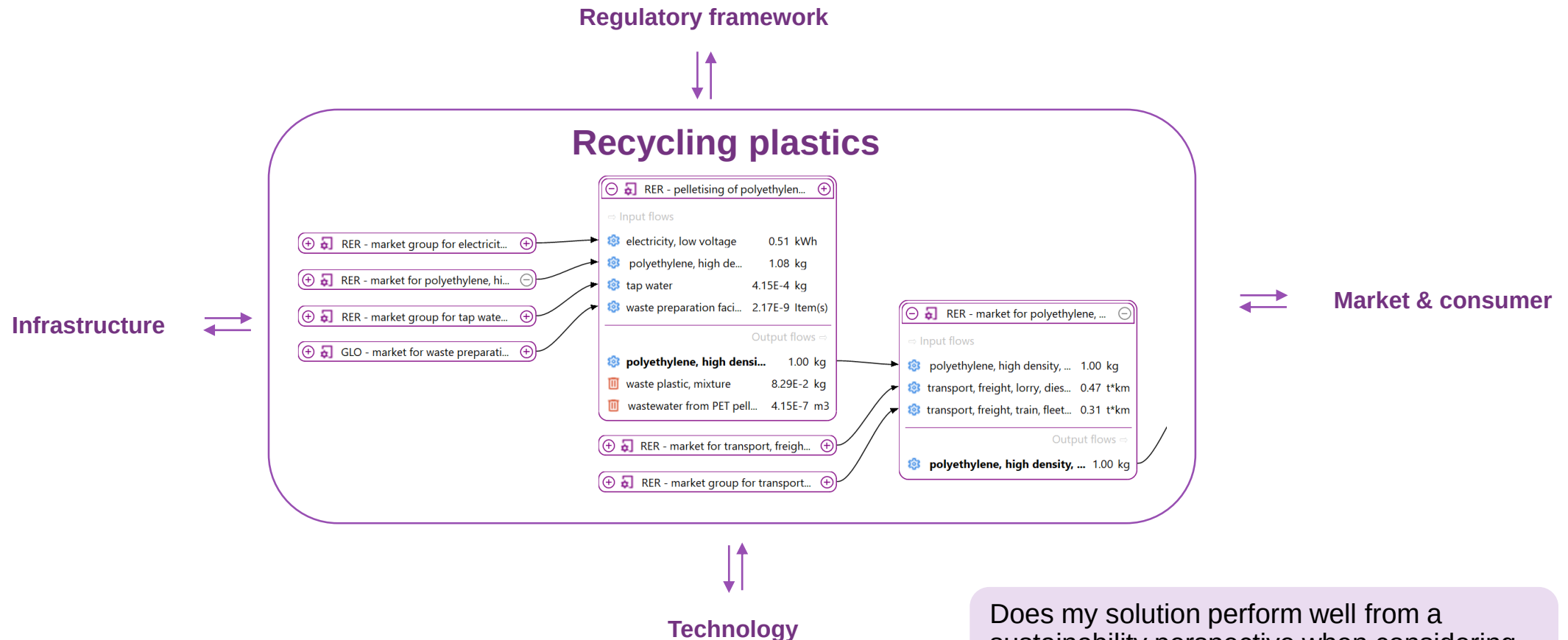
Limits to LCA?

What about: Thresholds?

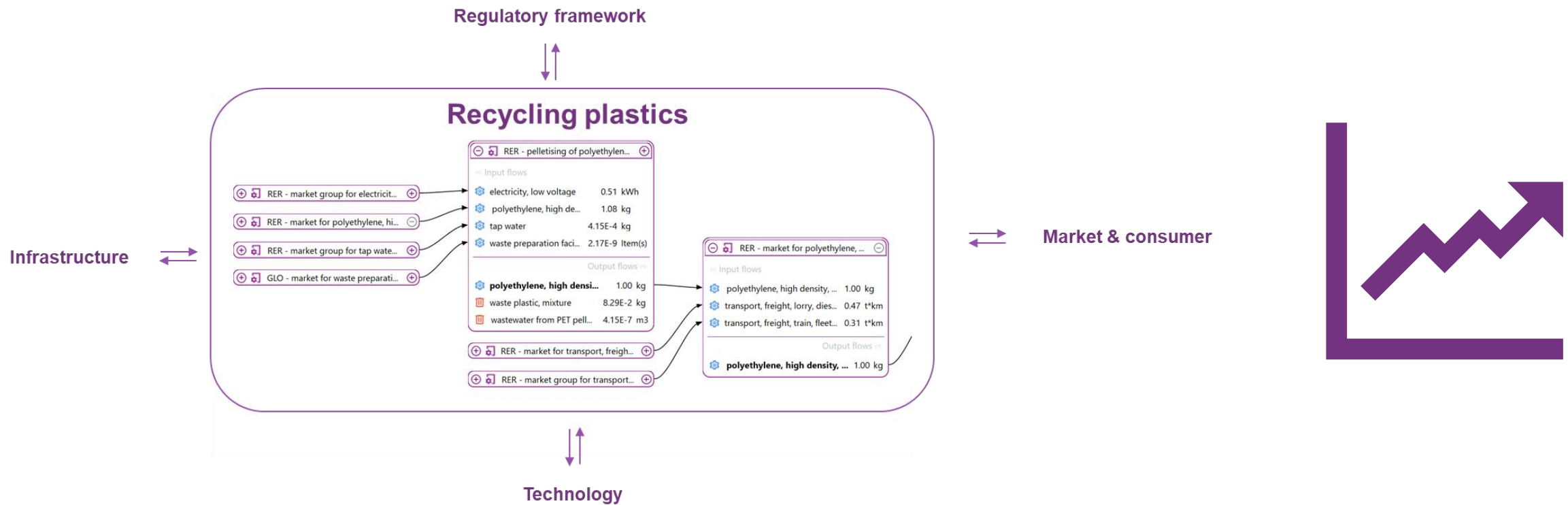


Does my solution operate within planetary boundaries?

What about: Context?



What about: Prospective view?

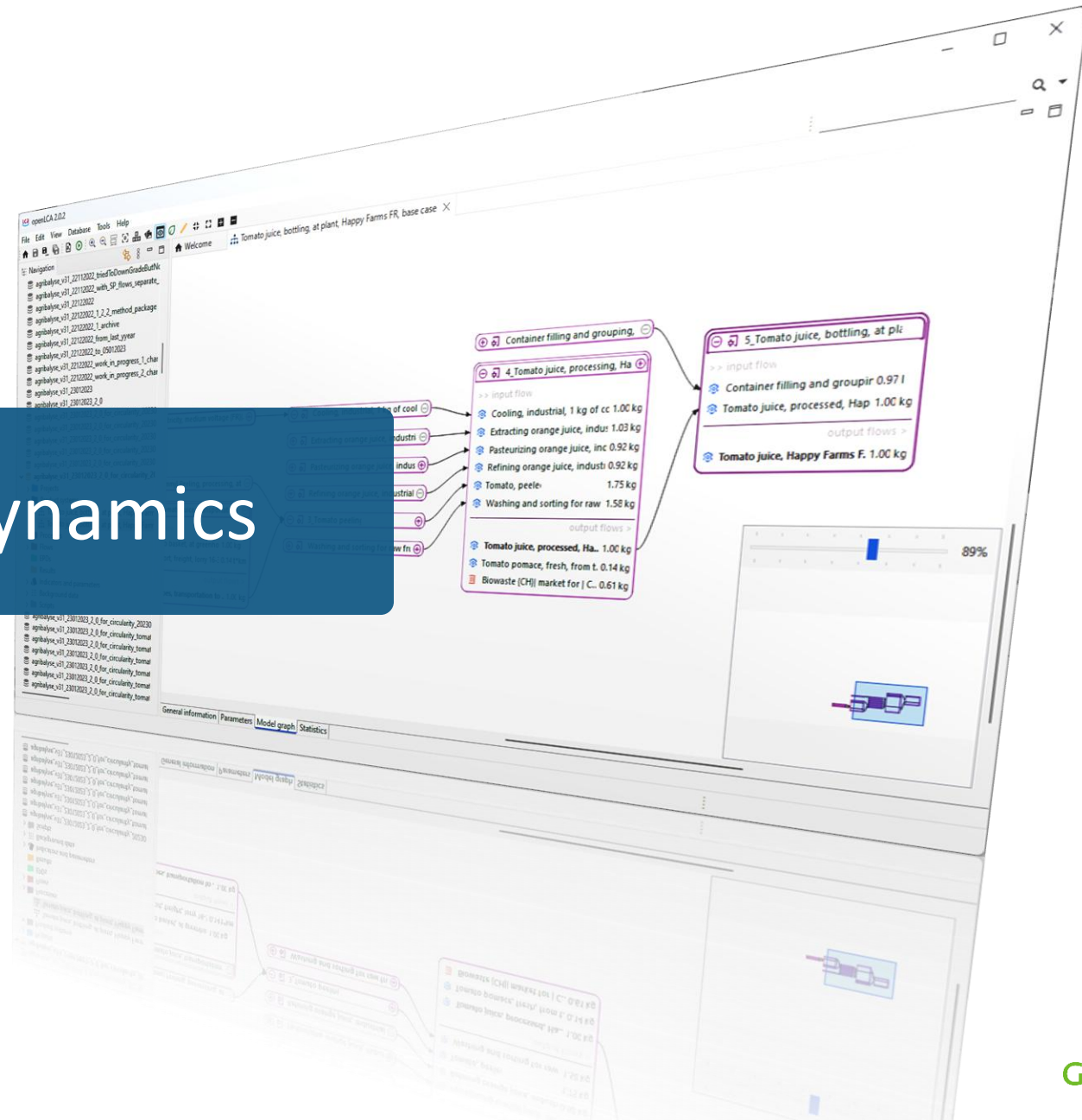


Does my solution perform well over time?

LCA today

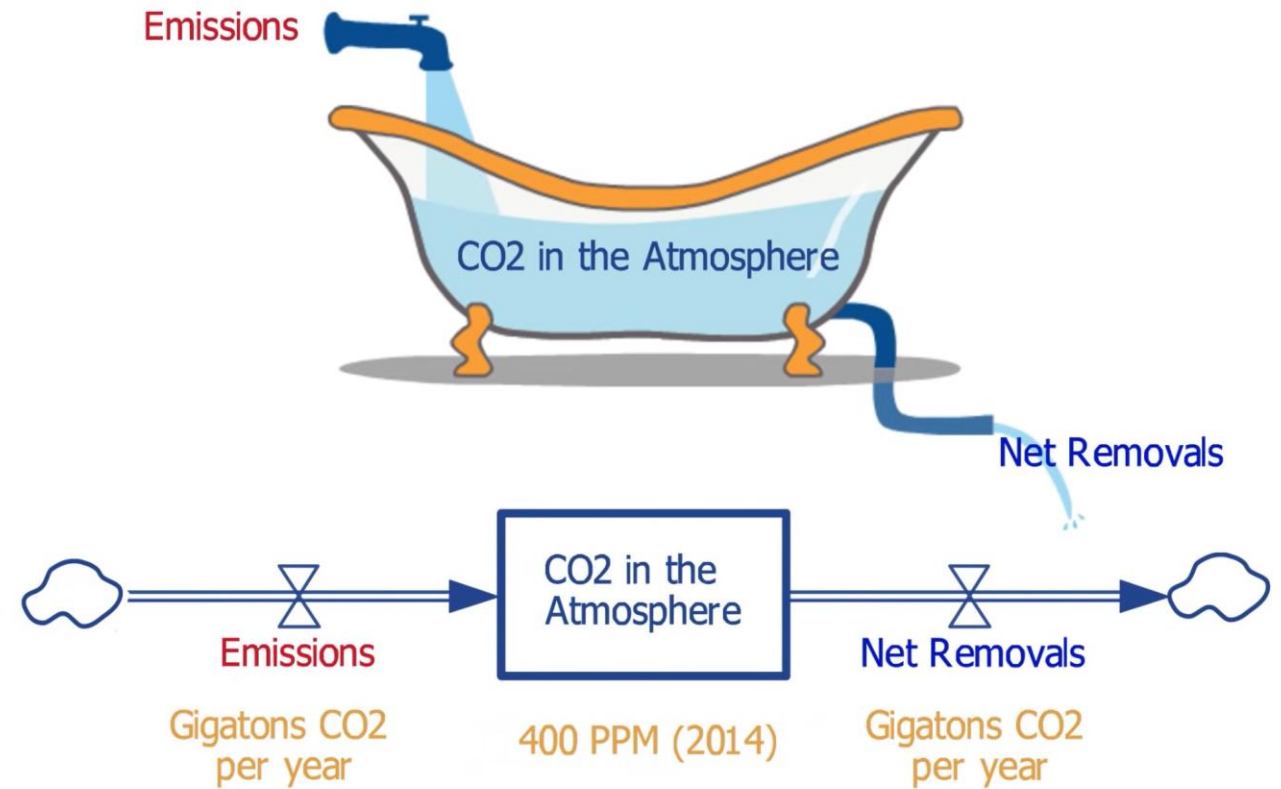
Capability	Absolute sustainability assessment	Consequential LCA	Dynamic LCA	Prospective LCA	System dynamics + LCA
Time-dependant impacts	X	X	✓	✓	✓
Future transition scenarios	X	X	✓	✓	✓
Market effects and indirect consequences	X	✓	Limited	Limited	✓
Feedback loops	X	X	X	X	✓
Socio-economic drivers and barriers	X	X	X	X	✓
Resource constraints	✓	X	X	X	✓
Planetary boundaries / thresholds	✓	X	X	X	✓

System Dynamics



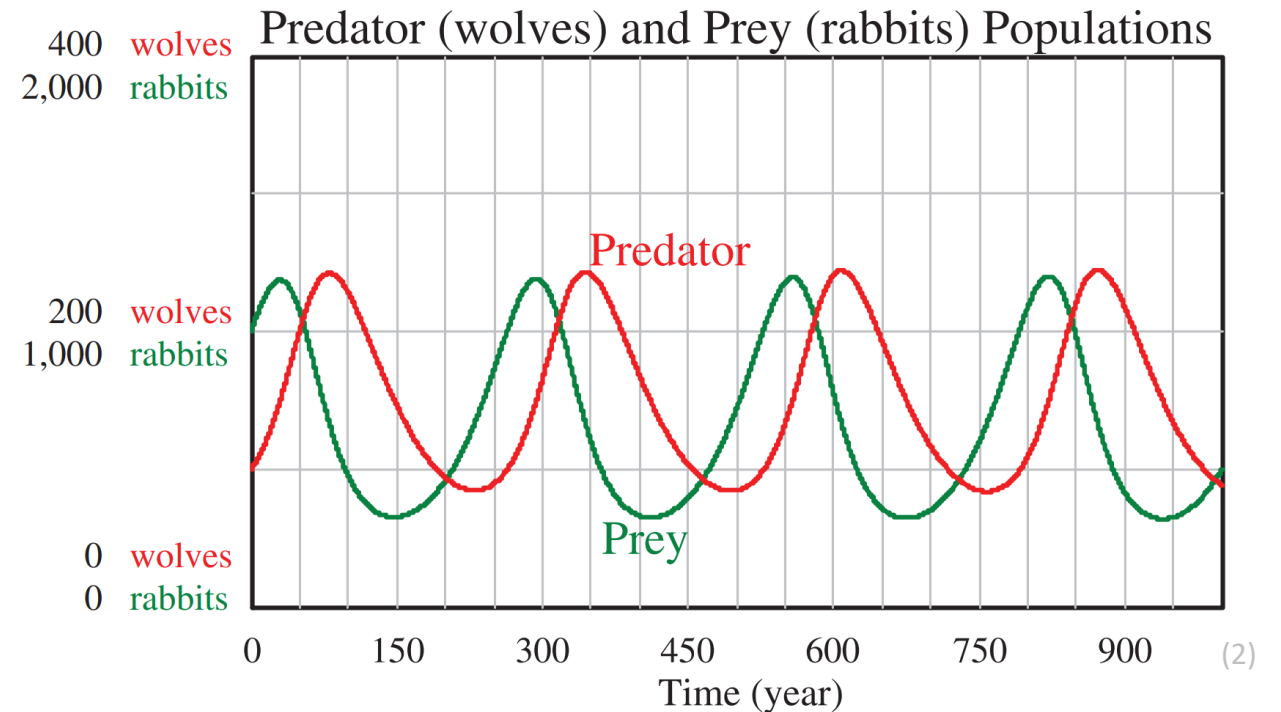
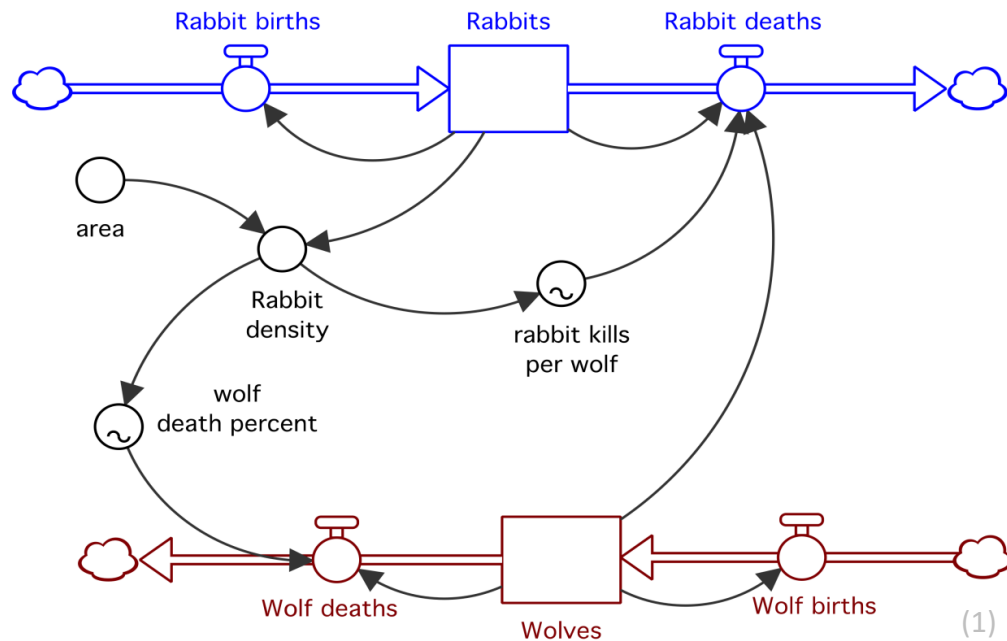
System dynamics

- Idea: modeling a system as a combination of stocks and flow rates
- Simulate results over time
- Any system(!)
- Scope and level of detail totally up to the modeler



System dynamics

- E.g., wolves and rabbits



(1) Fisher, Diana M. 2018. "Reflections on Teaching System Dynamics Modeling to Secondary School Students for over 20 Years" Systems 6, no. 2: 12.

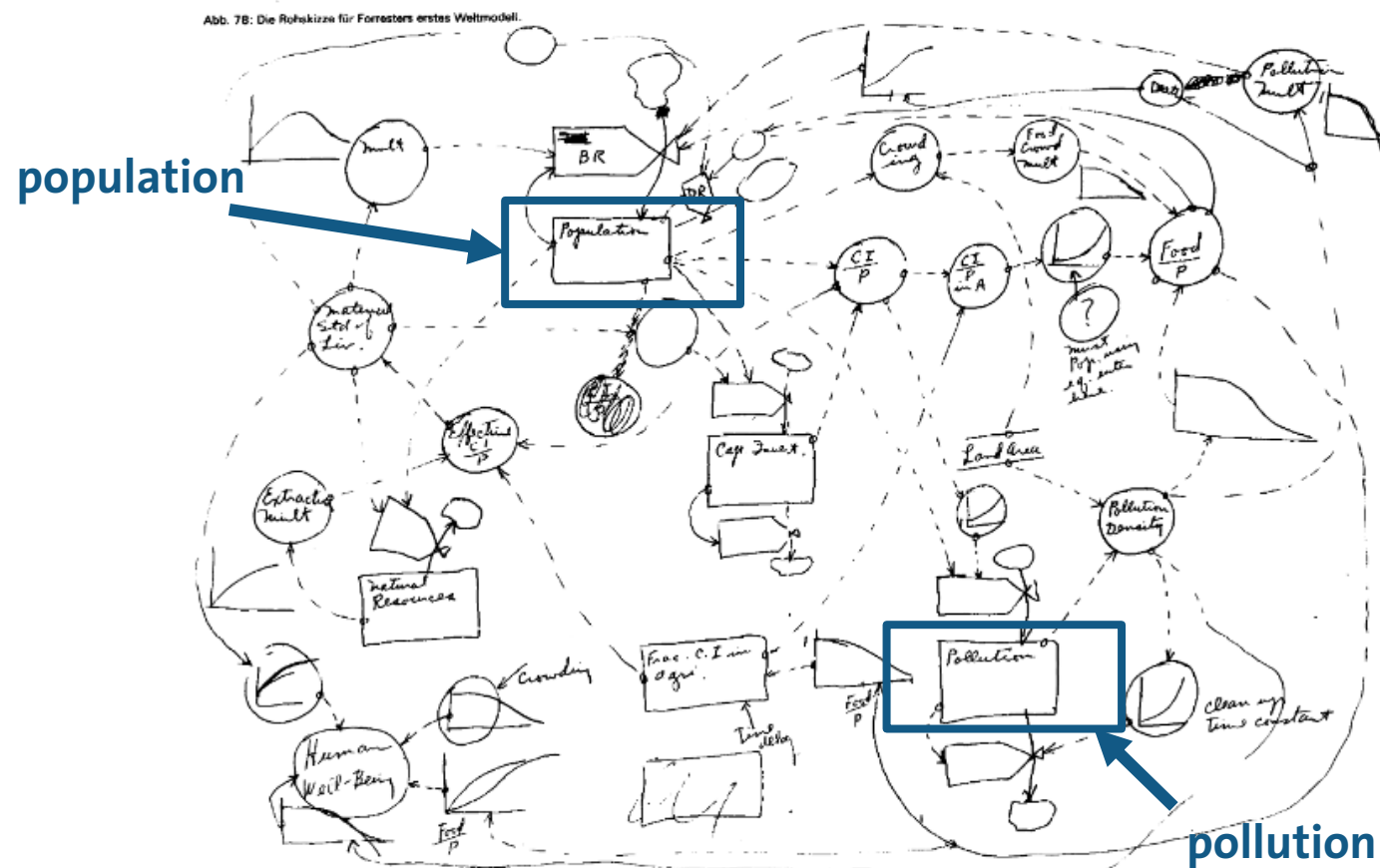
<https://doi.org/10.3390/systems6020012>

(2) Motesharrei, Safa & Rivas, Jorge & Kalnay, Eugenia. (2014). Human and nature dynamics (HANDY): Modeling inequality and use of resources in the collapse or sustainability of societies. Ecological Economics. 101. 90–102. 10.1016/j.ecolecon.2014.02.014.

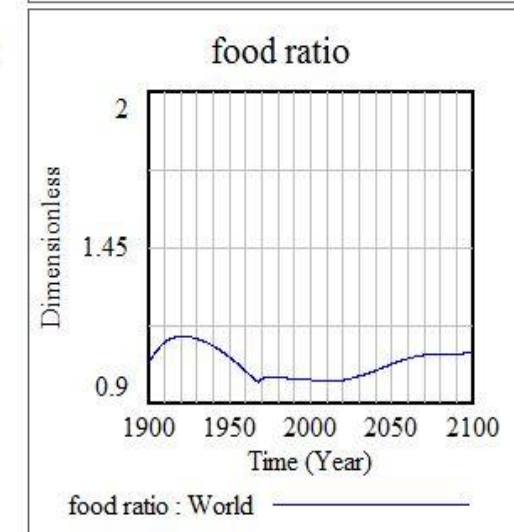
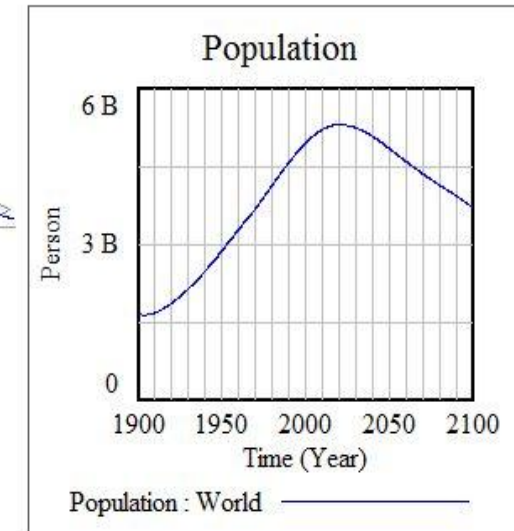
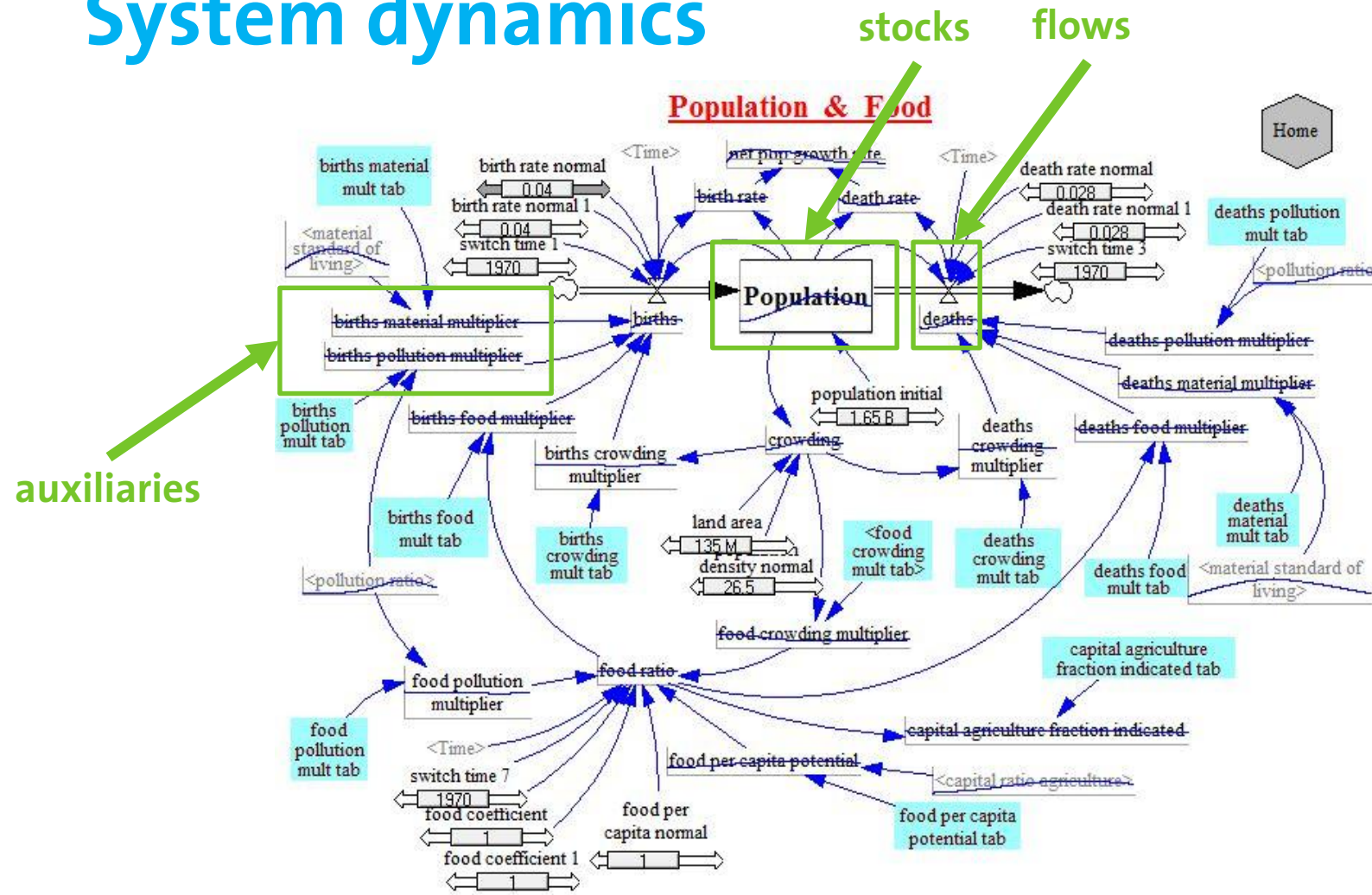
Slides taken from: Ciroth, A. (2023): Model – based LCSA, ISIE, Leiden, 2 July 2023.

System dynamics

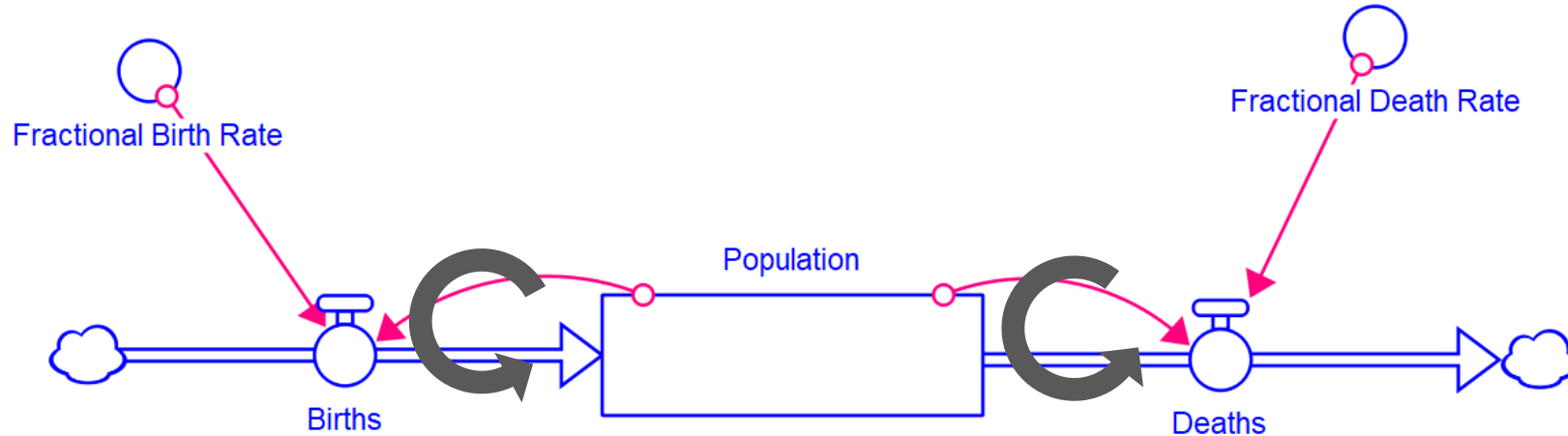
- Forrester et al. 1970's: a model of the entire world



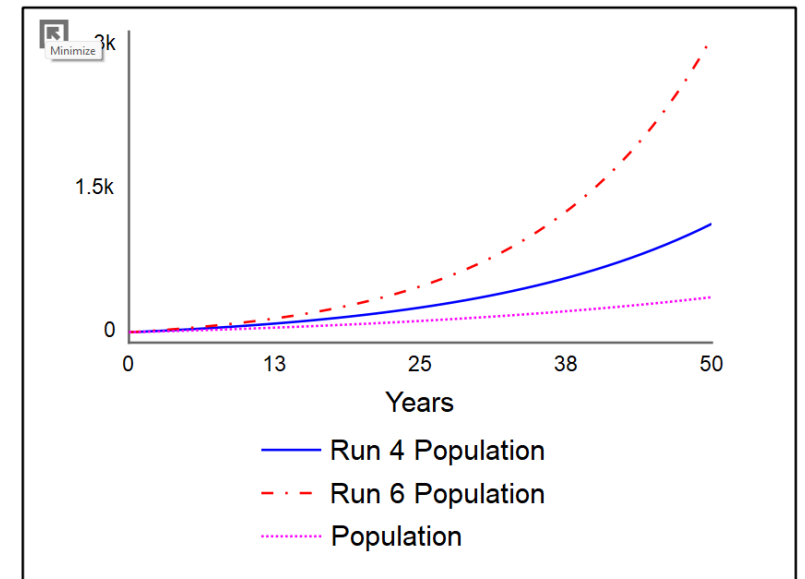
System dynamics



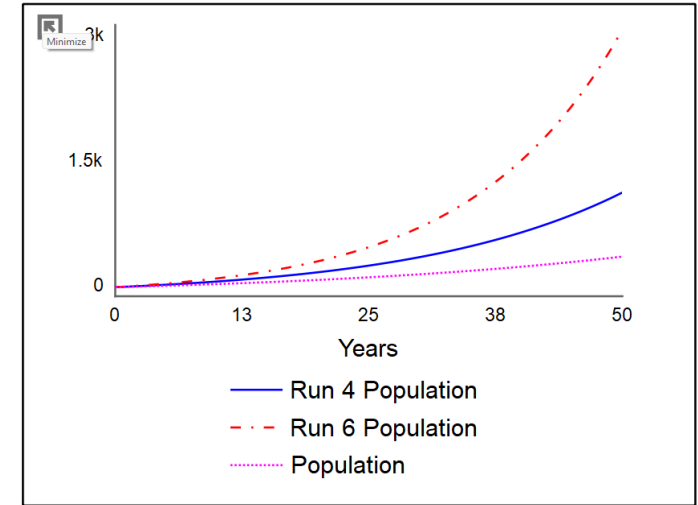
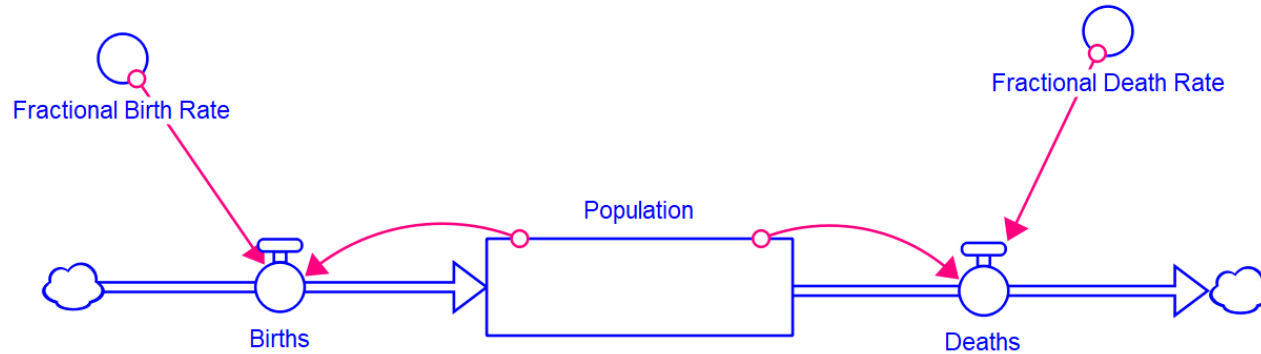
System dynamics



- Initial population = 100
- Births = Population * birth rate
- Deaths = Population * death rate
- Birth rate = 0.1
- Death rate = 0.05
- Reinforcing growth cycles



System dynamics



1. Calculate flow rates at the current time step – Using the current stock values and model equations, determine the inflows and outflows (or the derivative).
2. Apply Euler's update formula – Estimate the stock change over a time interval (Δt) as:

$$\text{Change} = \text{Net Flow} \times \Delta t$$

3. Update the stocks – Compute the next stock value by adding the estimated change to the current stock value:

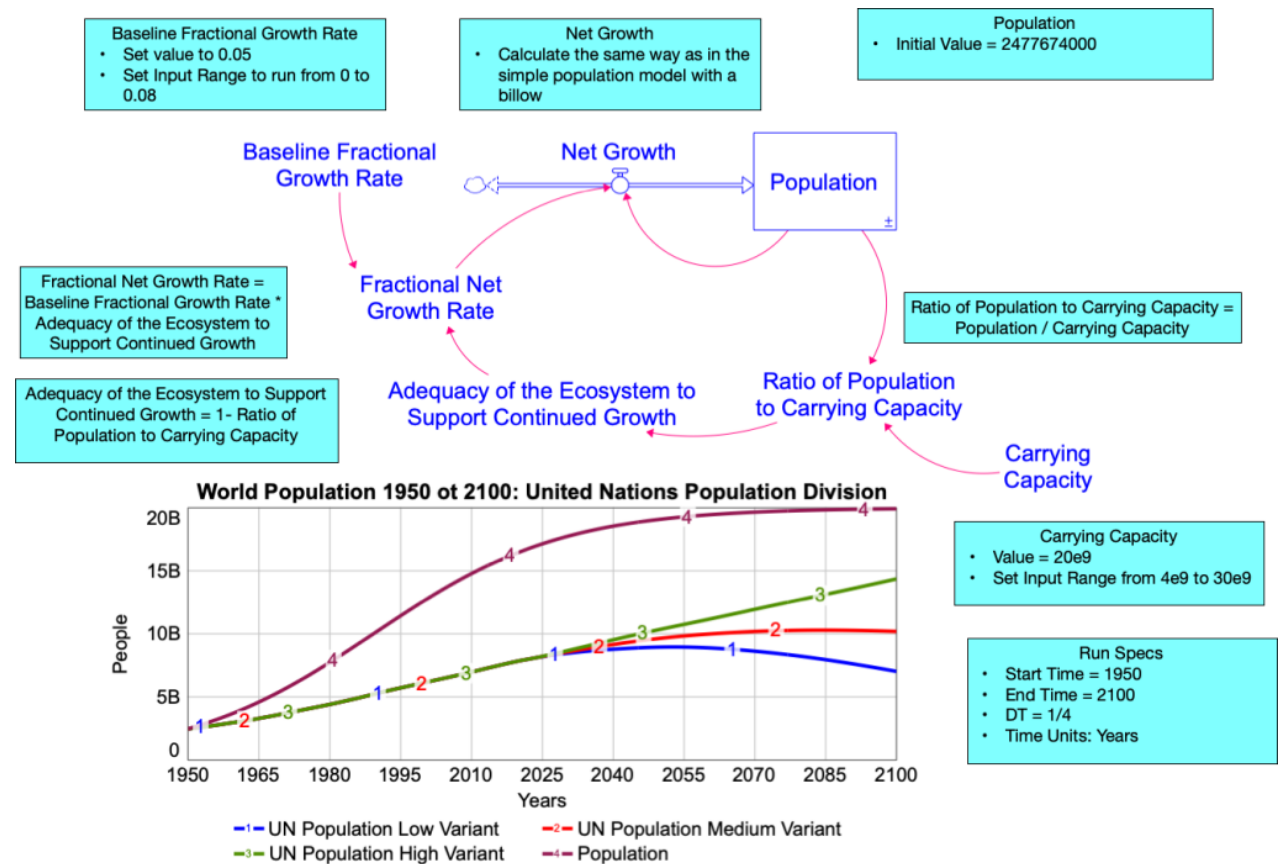
$$X_{t+\Delta t} = X_t + \left(\frac{dX}{dt} \right)_t \Delta t$$

4. Repeat for each time step

	Population	Births	Deaths	Fractional Birth Rate	Fractional Death Rate
0	100	10	5	0.1	0.05
1	105	10.5	5.25	0.1	0.05
2	110	11	5.51	0.1	0.05
3	116	11.6	5.79	0.1	0.05
4	122	12.2	6.08	0.1	0.05
5	128	12.8	6.38	0.1	0.05
6	134	13.4	6.7	0.1	0.05
7	141	14.1	7.04	0.1	0.05
8	148	14.8	7.39	0.1	0.05

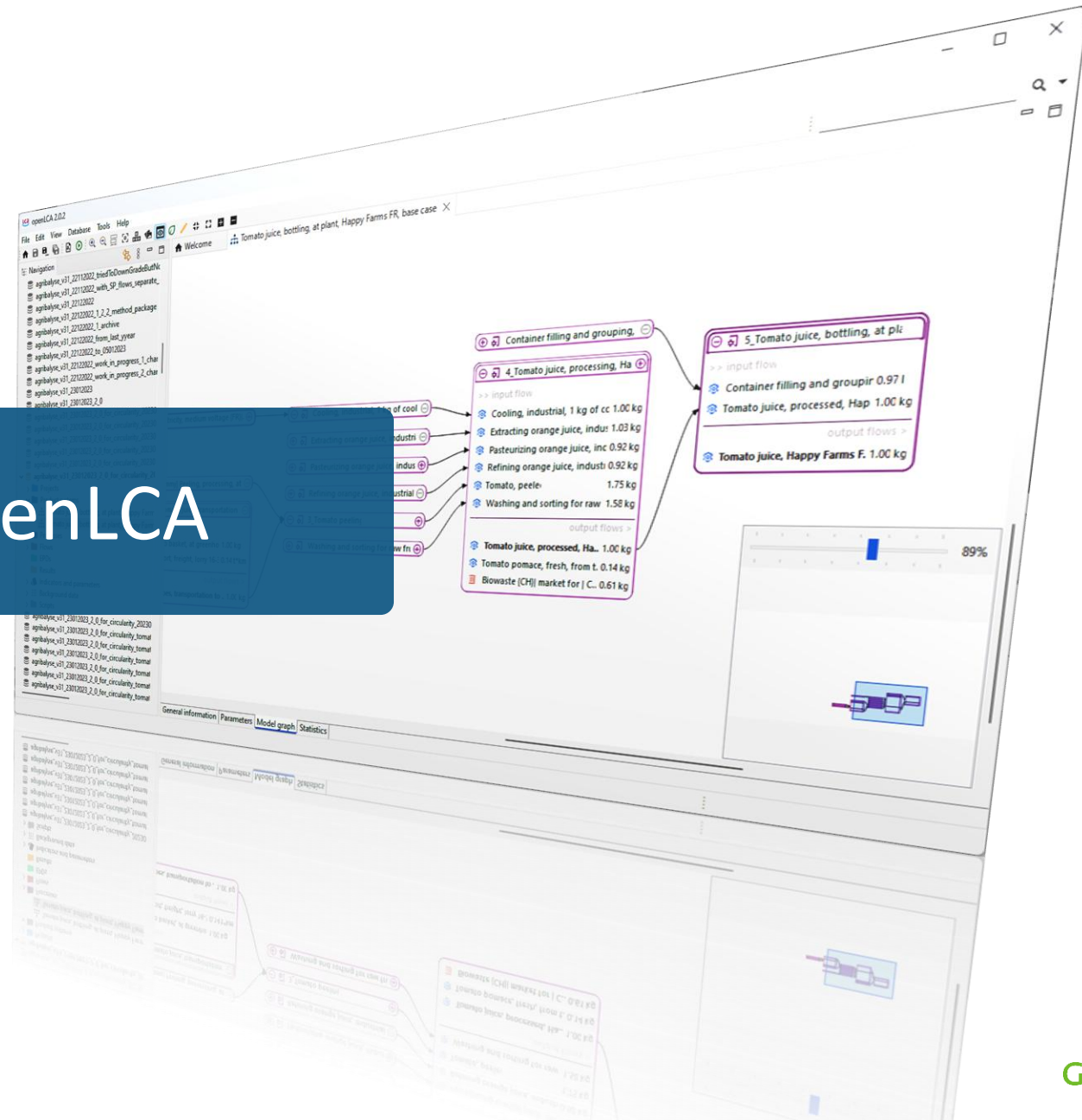
System dynamics

- Extending system dynamics models
- Using thresholds
- Using built-in functions (e.g. MAX; MIN; IF THEN ELSE)



Why is it useful to link SD and LCA?

SD in openLCA



System dynamics in openLCA

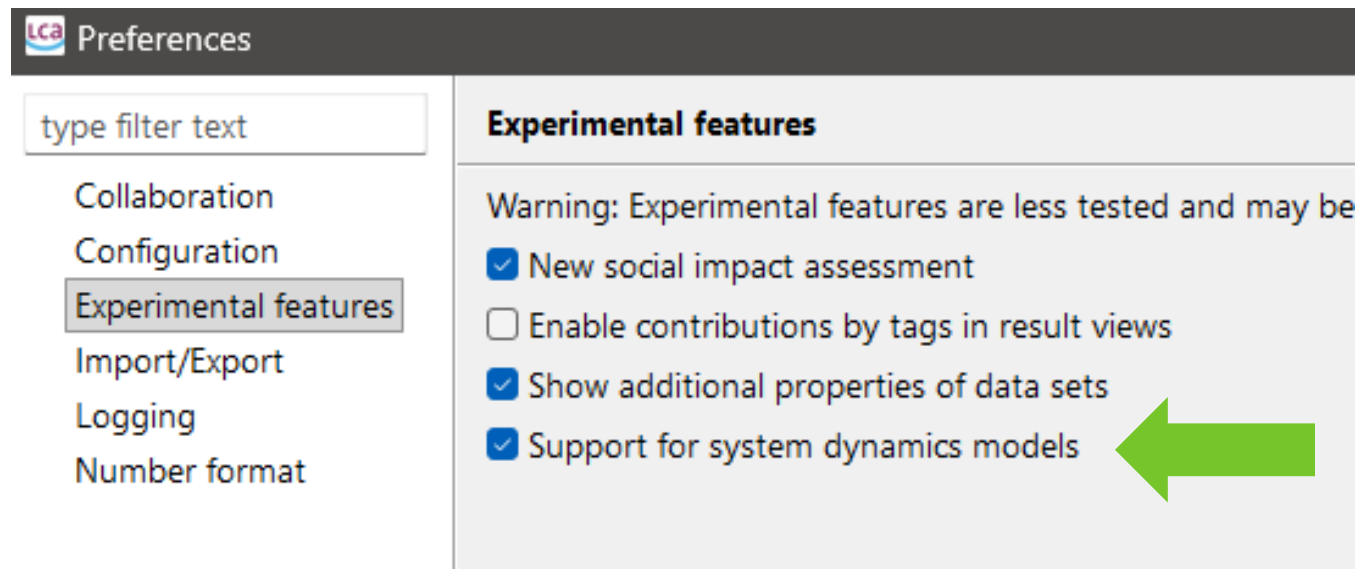
- Experimental feature since beginning of 2026
- An interpreter for system dynamics (SD) models. The idea is that SD models can be directly created, loaded and executed in openLCA.
- This is a first step to explore a possible SD integration in openLCA.

Basic functionality

- Following the XMILE standard
- Stocks
- Flows
- Auxiliaries
- Stand-alone graphical functions
- Various mathematical, statistical and time functions
- Array can be read but not directly edited (yet)

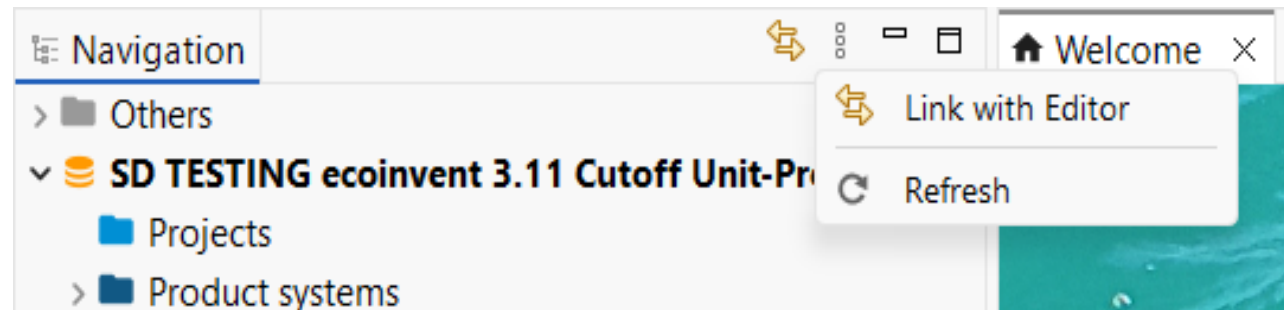
Enabling SD in openLCA

- First, you need to enable "Support for system dynamics models" in the preferences (do not forget to press "Apply" or "Apply & Close" in that dialog):



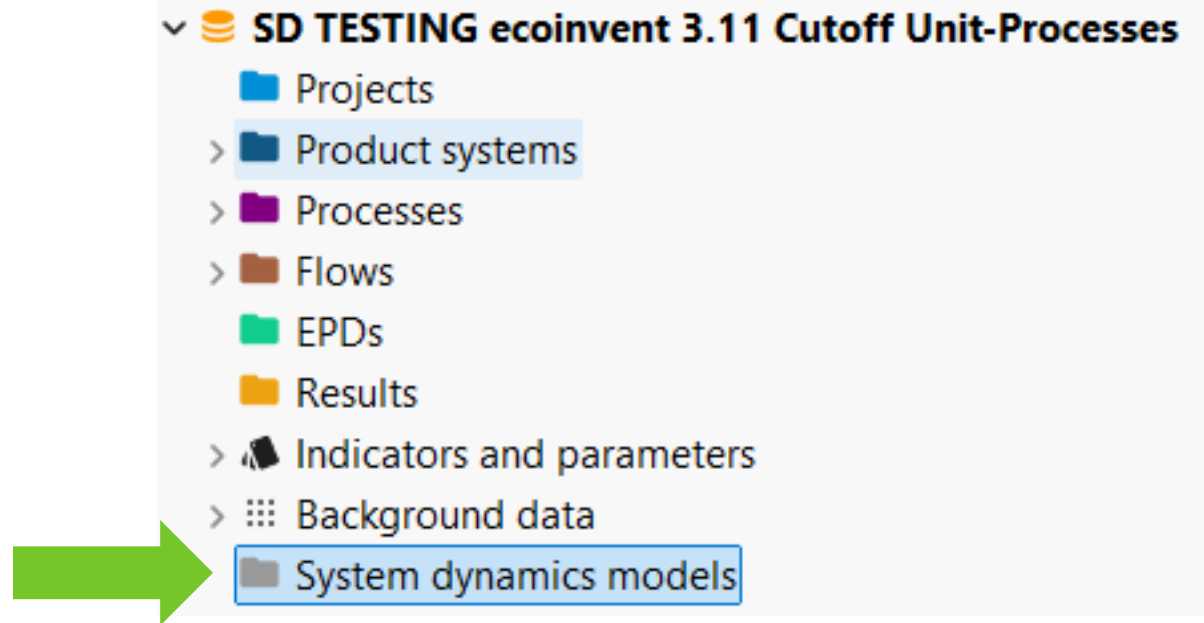
Enabling SD in openLCA

- Then, refresh the navigation via F5 or the refresh button:



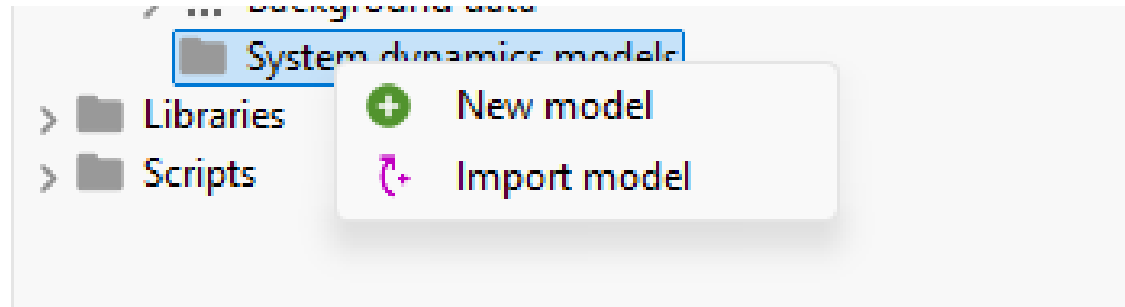
Enabling SD in openLCA

- You should see a folder "System dynamics models" in your database folders now:



SD models

- Right click to create a new model or import a model:



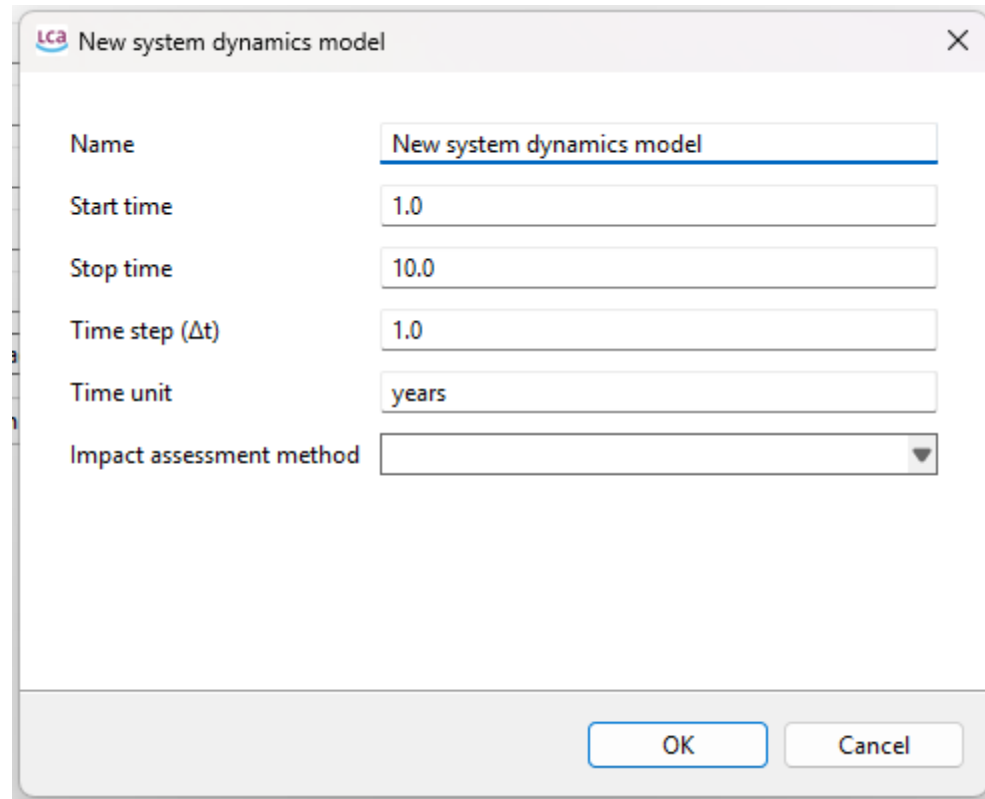
- Creating a new model opens a pop-up window, followed by an editor window
- Importing a model allows a user to import a SD model in the XMILE file format

XMILE

- A XMILE file contains information about a whole-model, with a well-specified structure.

```
<?xml version="1.0" encoding="utf-8"?>
<xmile version="1.0" xmlns="http://docs.oasis-open.org/xmile/ns/XMILE/v1.0" xmlns:isee="http://iseesystems.com/XMILE">
  <header>
    <smile version="1.0" namespace="std, isee" uses_arrays="2"/>
    <name>TREASoURcE - Plastic substitution model v7.1 - with KVC</name>
    <uuid>eed773d1-f4f7-4725-8d05-452b96d4f2dc</uuid>
    <vendor>isee systems, inc.</vendor>
    <product version="3.5" isee:build_number="3112" isee:saved_by_v1="true" lang="en">Stella Architect</product>
  </header>
  <sim_specs isee:sim_duration="1.5" isee:run_prefix="Run" isee:simulation_delay="0.06" isee:restore_on_start="false" method="Euler" time_units="Years"
isee:instantaneous_flows="false" isee:ignore_module_errors="false" isee:strict_units="true" isee:loop_scores="true" isee:loop_exhaustive_allowed="1000">
    <start>0</start>
    <stop>25</stop>
    <dt reciprocal="true">1</dt>
  </sim_specs>
  <dimensions>
    <dim name="Products">
      <elem name="Plastics_primary"/>
      <elem name="Plastic_recycled"/>
    </dim>
    <dim name="Waste">
      <elem name="Plastic_waste"/>
    </dim>
  </dimensions>
  <isee:prefs saved_runs="5" keep="false" show_module_prefix="true" live_update_on_drag="true" show_restore_buttons="false" layer="model"
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    <isee:time_format name="Builtin" type="adaptive"/>
  </isee:time_formats>
  <isee:number_format_extenders/>
</xmile>
```

Creating a new SD model



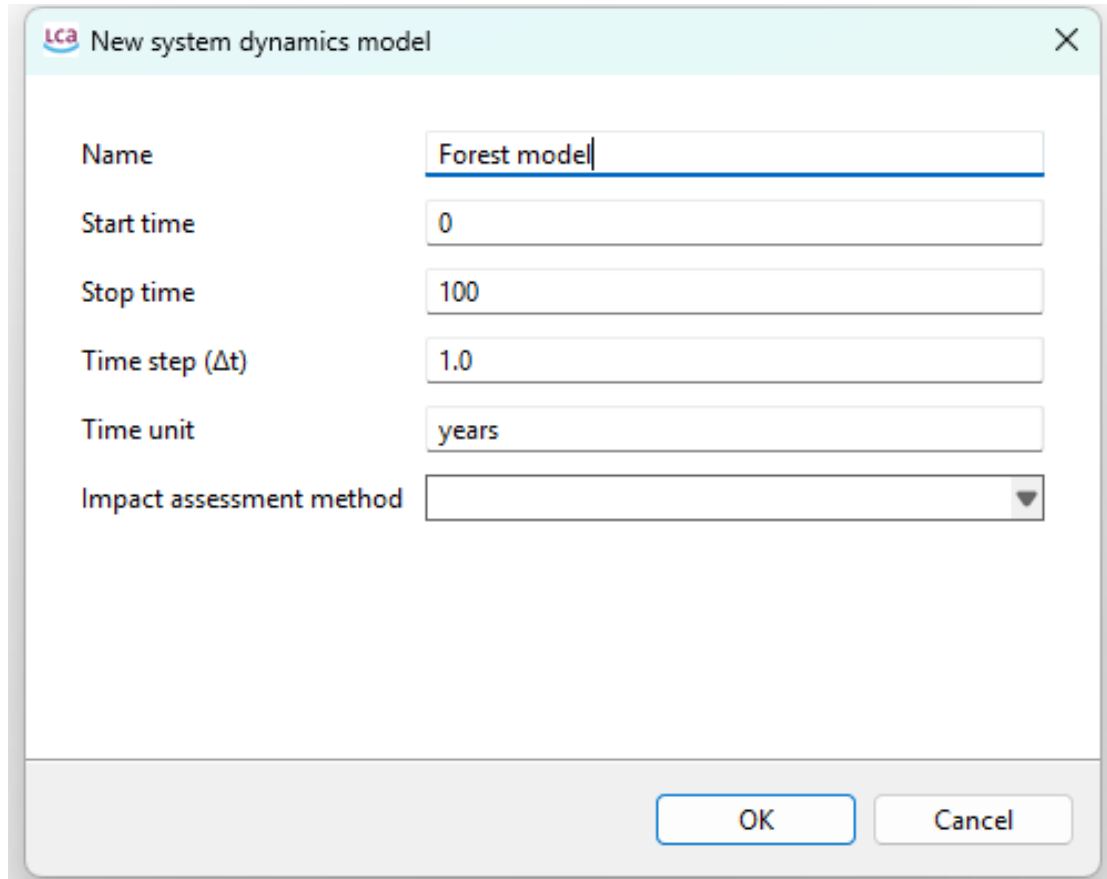
The screenshot shows a dialog box titled "New system dynamics model" with a close button (X) in the top right corner. The dialog contains several input fields and a dropdown menu. The "Name" field is pre-filled with "New system dynamics model". The "Start time" field is set to "1.0", the "Stop time" field is set to "10.0", and the "Time step (Δt)" field is set to "1.0". The "Time unit" field is set to "years". The "Impact assessment method" field is a dropdown menu with a downward arrow. At the bottom of the dialog are "OK" and "Cancel" buttons.

Field	Value
Name	New system dynamics model
Start time	1.0
Stop time	10.0
Time step (Δt)	1.0
Time unit	years
Impact assessment method	

Information

- Name of your SD model
- Start time for your SD simulation
- Stop time for your SD simulation
- Time steps (delta time)
- Time unit (days, months, years, ...)
- Life cycle impact assessment method

Exercise 1: Creating a new SD model



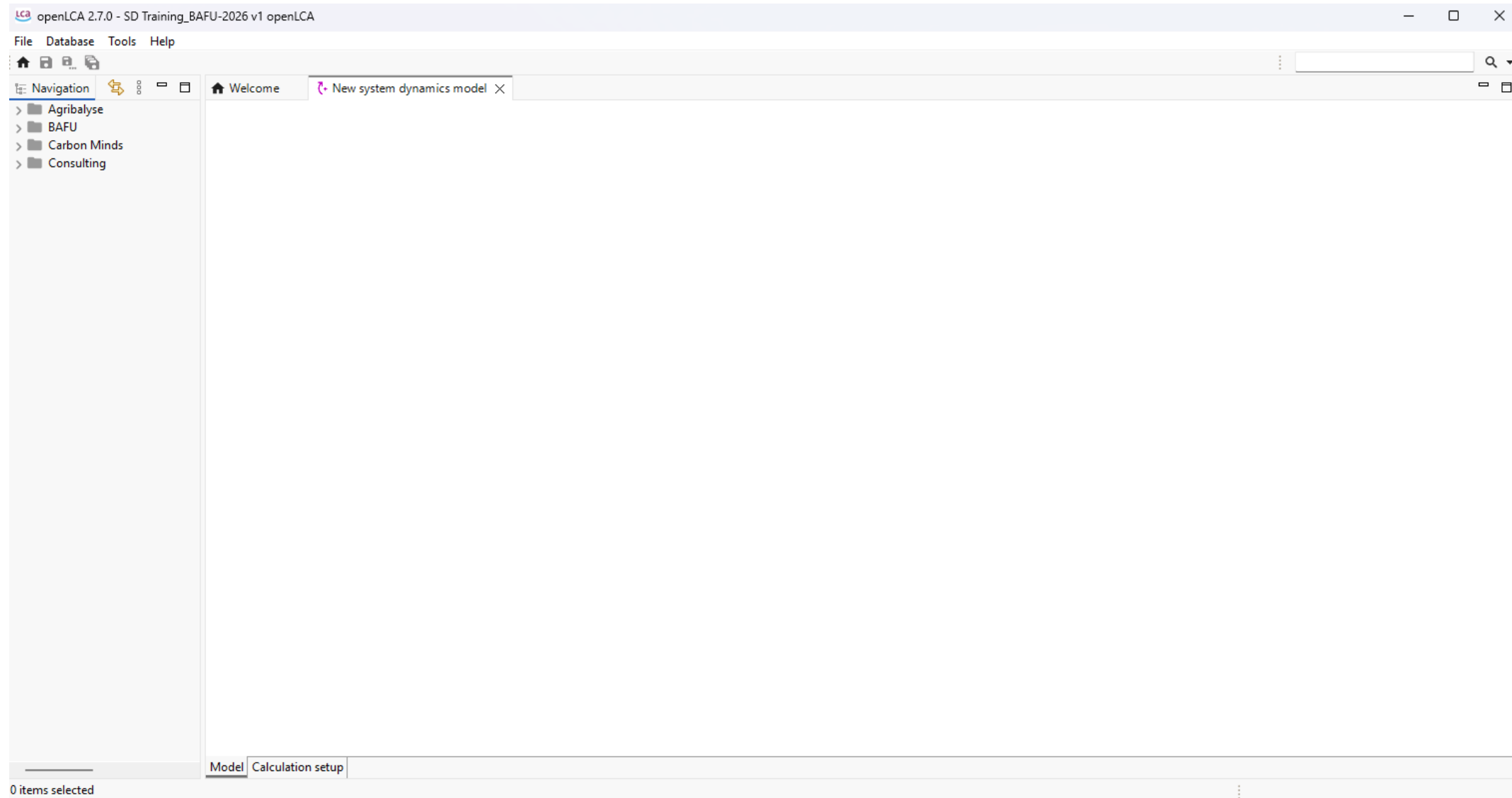
The screenshot shows a dialog box titled "New system dynamics model" with a close button (X) in the top right corner. The dialog contains several input fields and a dropdown menu. The "Name" field is filled with "Forest model". The "Start time" field is filled with "0". The "Stop time" field is filled with "100". The "Time step (Δt)" field is filled with "1.0". The "Time unit" field is filled with "years". The "Impact assessment method" field is a dropdown menu with a downward arrow. At the bottom of the dialog are two buttons: "OK" and "Cancel".

Field	Value
Name	Forest model
Start time	0
Stop time	100
Time step (Δt)	1.0
Time unit	years
Impact assessment method	

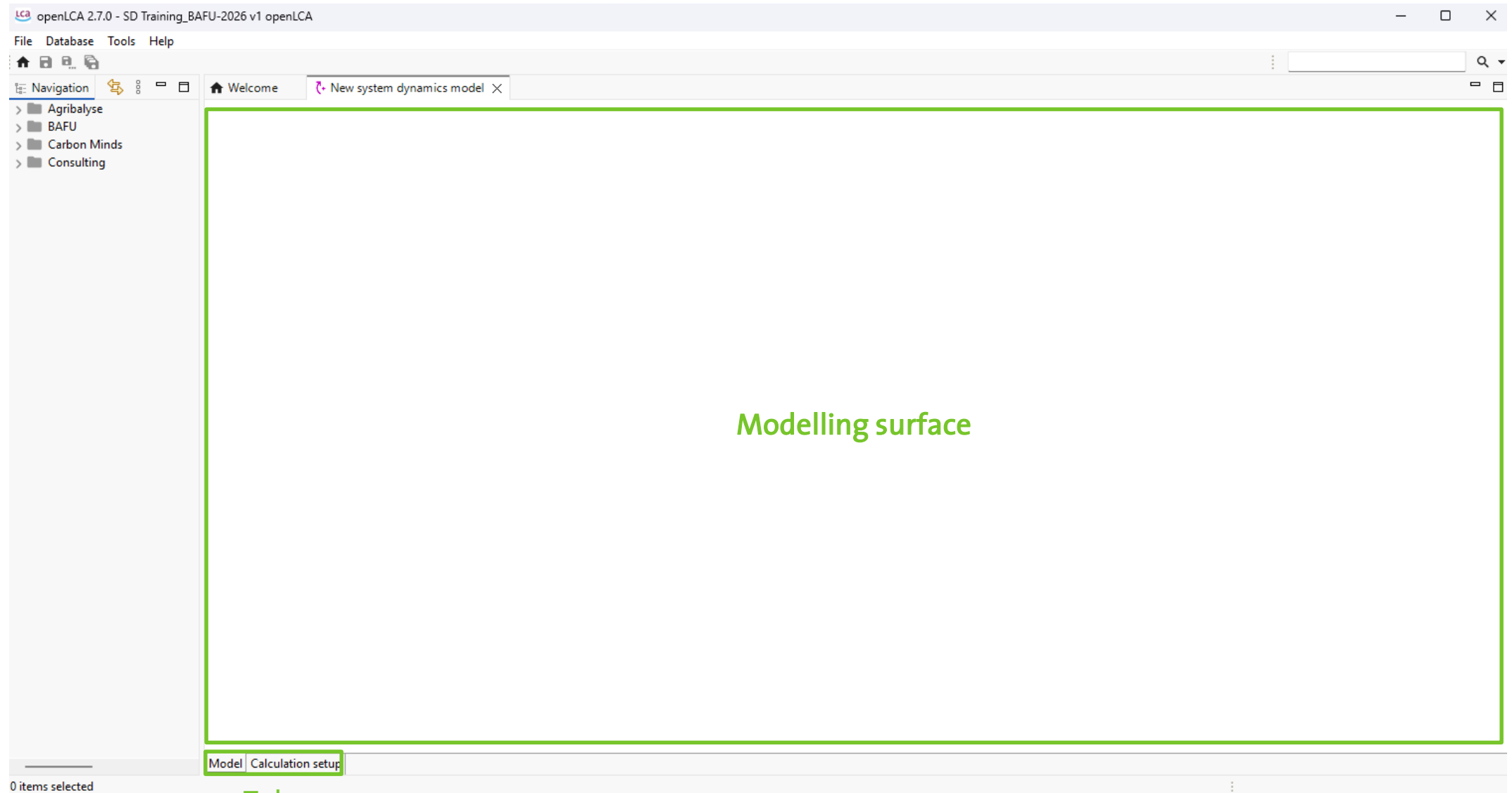
Case study example

- A model that looks at a forest and explores different wood production pathways
- To observe effects: timeframe of 100 years
- Time steps: 1
- No impact assessment method (for now)

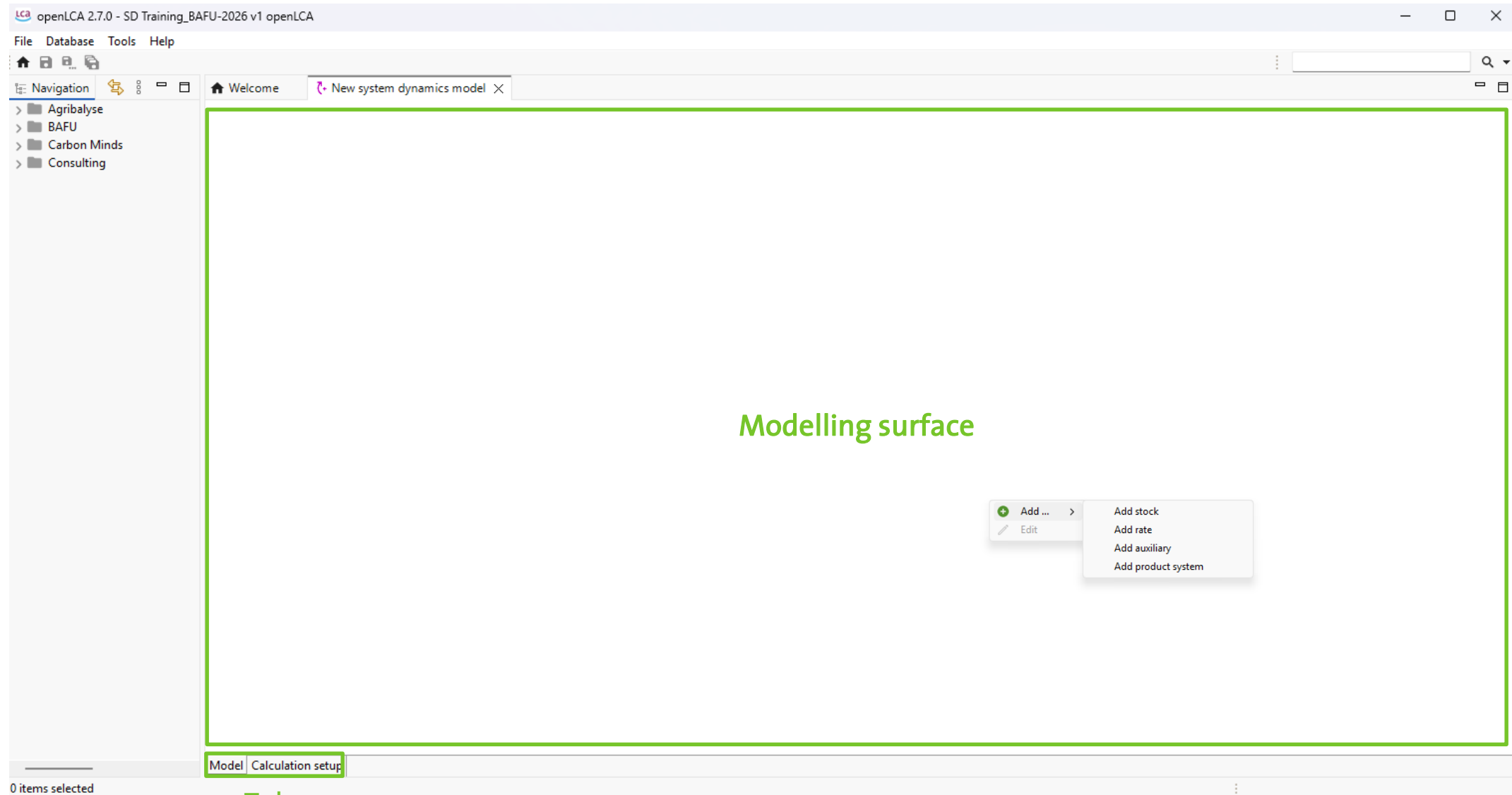
Creating a new SD model



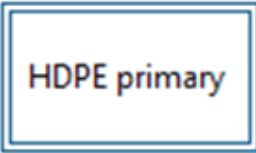
Creating a new SD model



Creating a new SD model



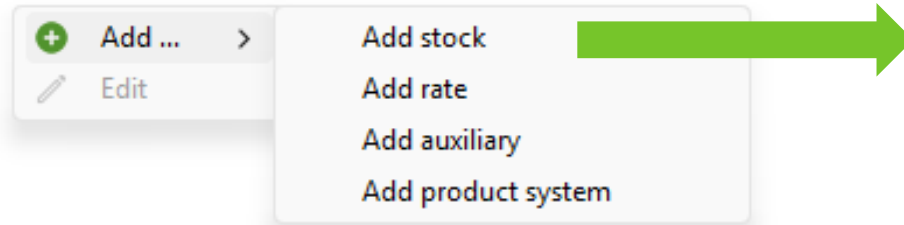
SD elements in openLCA

Stock	Flow	Auxiliary	Product system
			

The system dynamics elements are defined as follows:

- Stocks – are accumulations or state variables in a system that represent quantities at a given point in time (e.g., population, inventory, or savings). They change only through flows.
- Flows – are the rates at which stocks increase or decrease over time (e.g., births, sales, or withdrawals). They determine how stocks evolve.
- Auxiliary – are intermediate variables used to calculate flows or provide additional logic and relationships in the model (e.g., birth rate, demand factor, or policy settings). They do not accumulate over time themselves.

Stocks



A screenshot of a dialog box titled 'LCA'. The dialog box has a title bar with a close button (X). The main content area contains the following fields and controls:

- Name:** A text input field.
- Unit:** A text input field.
- Non-negative:** A checkbox.
- Input flows:** A text input field.
- Output flows:** A text input field.
- Type:** A dropdown menu with 'Equation' selected.
- Equation:** A large text area for entering the equation.

At the bottom right, there are two buttons: 'OK' and 'Cancel'.

Stocks

Adding input and output flows

The dialog box is titled 'LCA' and has a close button (X) in the top right corner. It contains the following fields and controls:

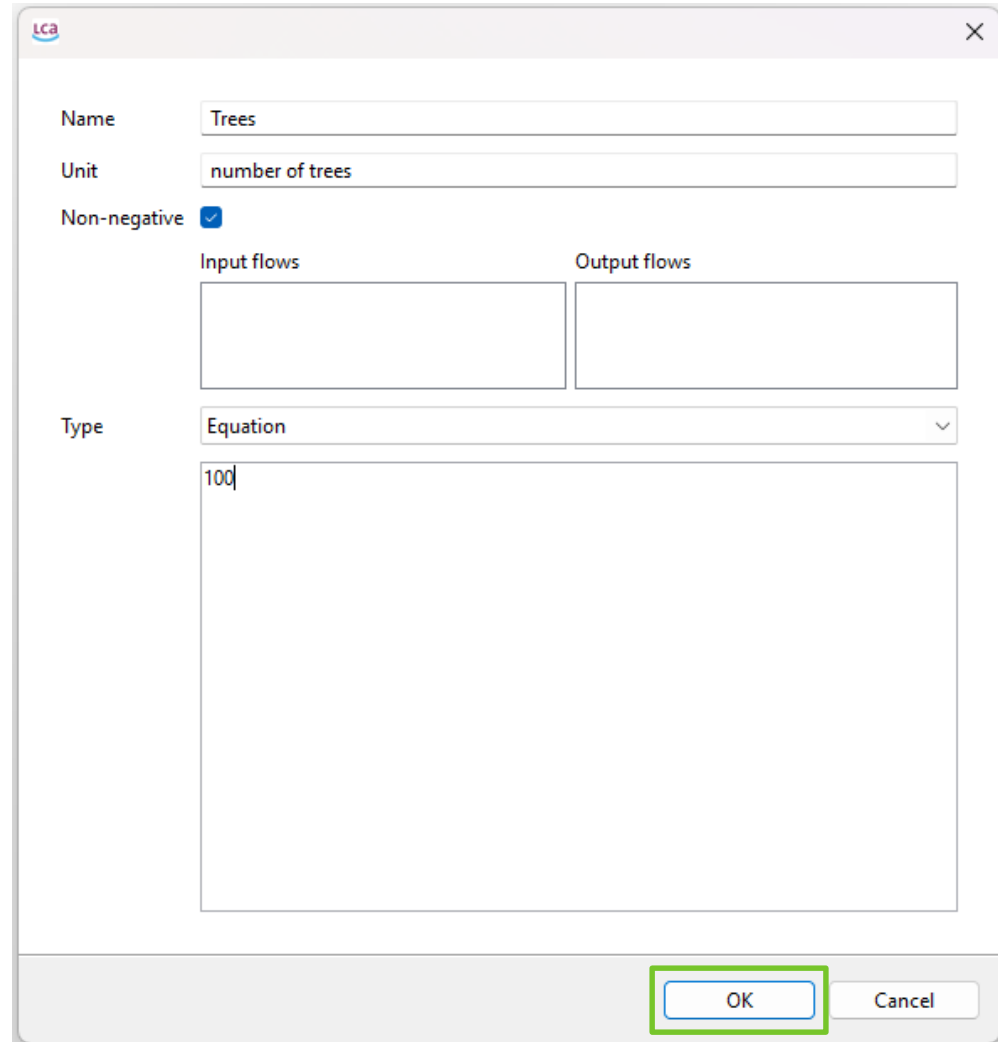
- Name:** A text input field.
- Unit:** A text input field.
- Non-negative:** A checkbox.
- Input flows:** A text input field with a small floating button containing a green '+' icon and the text 'Add', and a red 'X' icon with the text 'Remove'.
- Output flows:** A text input field.
- Type:** A dropdown menu currently showing 'Equation'.
- Equation editor:** A large text area for entering the equation.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Choosing stock type

The dialog box is titled 'LCA' and has a close button (X) in the top right corner. It contains the following fields and controls:

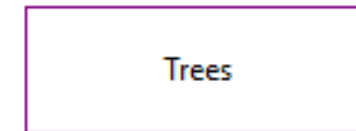
- Name:** A text input field.
- Unit:** A text input field.
- Non-negative:** A checkbox.
- Input flows:** A text input field.
- Output flows:** A text input field.
- Type:** A dropdown menu with a list of options: 'Equation', 'Equation', 'Lookup function', and 'Array'. The first 'Equation' option is highlighted in blue.
- Equation editor:** A large text area for entering the equation.
- Buttons:** 'OK' and 'Cancel' buttons at the bottom right.

Stocks

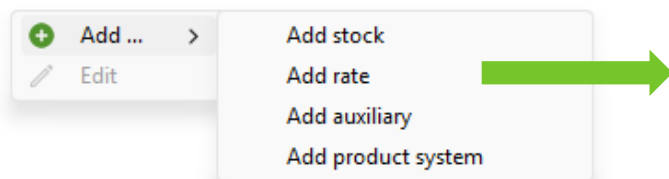


The screenshot shows a dialog box titled 'LCA' with a close button (X) in the top right corner. The dialog contains the following fields and controls:

- Name:** A text box containing the word 'Trees'.
- Unit:** A text box containing the text 'number of trees'.
- Non-negative:** A checkbox that is checked.
- Input flows:** An empty rectangular box.
- Output flows:** An empty rectangular box.
- Type:** A dropdown menu currently showing 'Equation'.
- Equation:** A large text area containing the number '100'.
- Buttons:** At the bottom right, there are two buttons: 'OK' and 'Cancel'. The 'OK' button is highlighted with a green rectangular border.



Rates



A dialog box titled "LCA" with a close button. It contains the following fields:

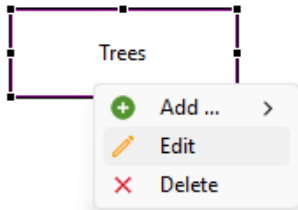
- Name: tree growth
- Unit: trees/year
- Non-negative: ☐
- Type: Equation (dropdown menu)
- Equation: 5

At the bottom are "OK" and "Cancel" buttons.



Connecting rates and stocks

tree growth



LCA

Name: Trees

Unit: number of trees

Non-negative: ☒

Input flows: + Add ✗ Remove

Output flows:

Type: Equation

100

OK Cancel

LCA Select flow(s)

Filter

tree growth

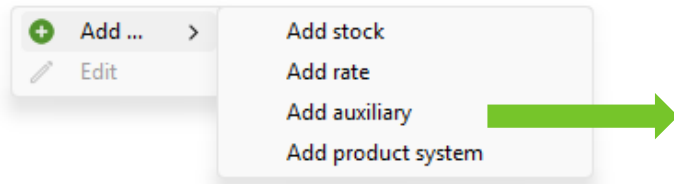
OK Cancel

tree growth



Trees

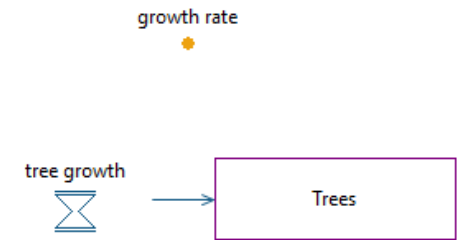
Auxiliaries



A dialog box titled "LCA" with a close button (X) in the top right corner. It contains the following fields and options:

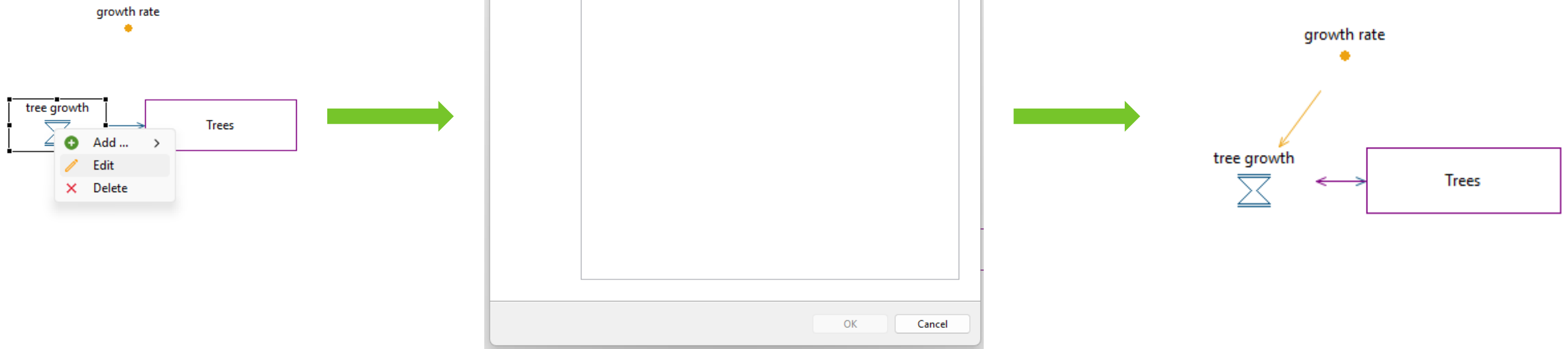
- Name:
- Unit:
- Non-negative: ☐
- Type:
- Equation:

At the bottom right are "OK" and "Cancel" buttons.



Model equations

Trees*growth_rate



Exponential growth function $\frac{dTrees}{dt} = r Trees$

Running the simulation

System dynamics model: New system dynamics model

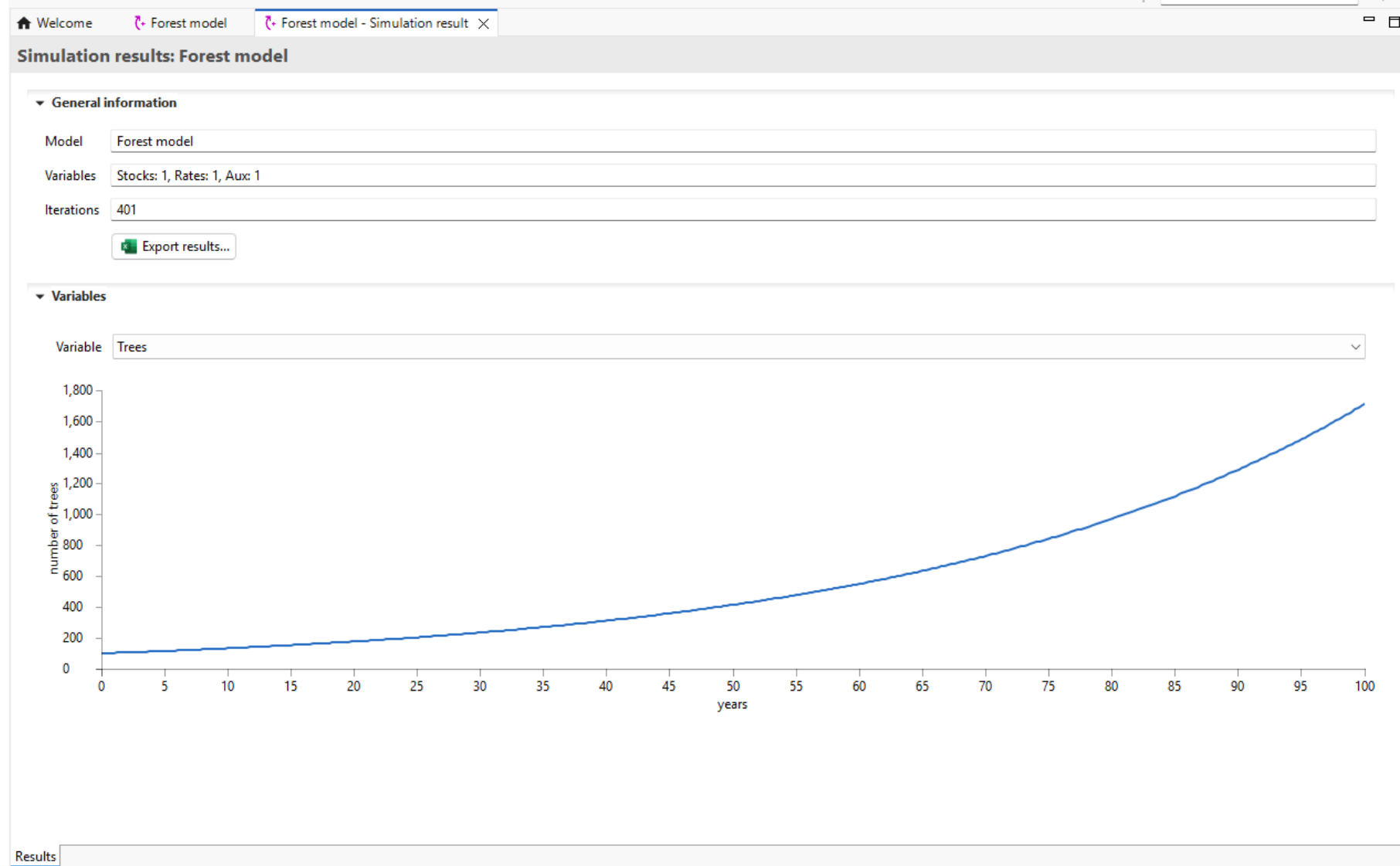
▼ General information

Name	Forest model	
Start time	0	years
Stop time	100	years
Δt	0.25	years
Time unit	years	
Impact assessment method		

 [Run simulation](#)

Model | Calculation setup

System dynamics results

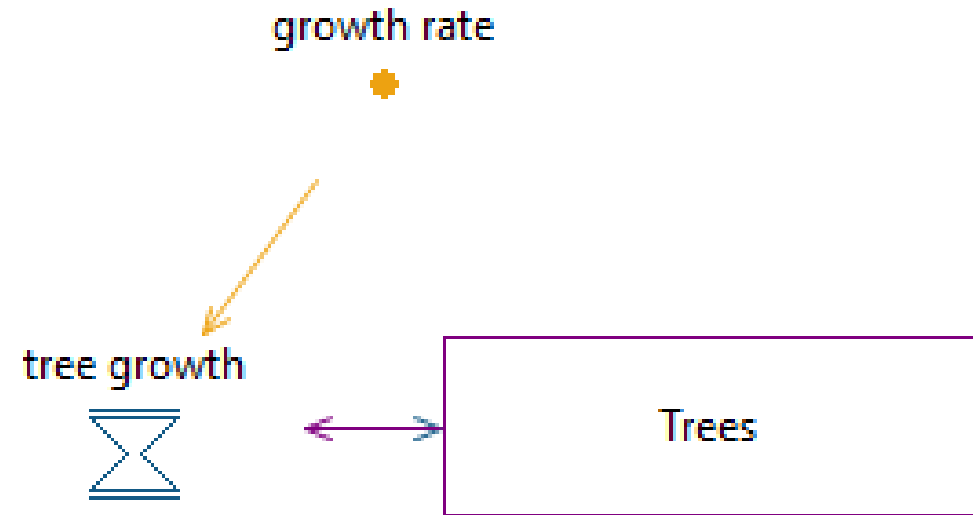


Exercise 2: Create a simple SD model

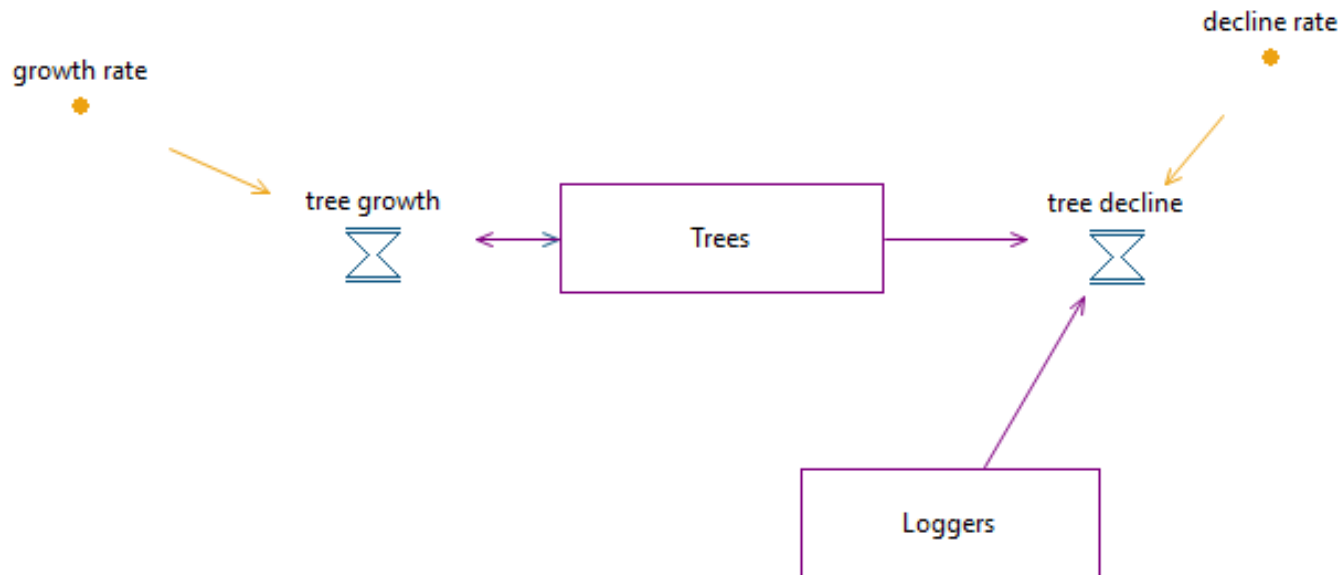
Case study assumption:

- Initial tree population = 90 Trees
- Growth rate = 8% / year
- Tree growth = Trees * growth rate

Run simulation: what can you see?



Exercise 3: Extending the model



Case study assumption:

- $\text{Loggers} = 3$
- $\text{Decline rate} = 0.5 \text{ trees/year}$
- $\text{Tree decline} = \text{Loggers} * \text{decline rate}$

Run simulation: what can you see?

Functions

Functions currently implemented in openLCA

- TIME → returns the current simulation time
- IF THEN ELSE → conditional function
- MAX → returns largest value
- MIN → returns smallest value

Other functions

ABS	INT	EXPRND	STOPTIME
ARCCOS	LN	LOGNORMAL	INIT
ARCSIN	LOG10	NORMAL	
ARCTAN	PI	POISSON	
COS	SIN	RANDOM	
EXP	SQRT	DT	
INF	TAN	STARTTIME	

Exercise 4: Extending the model

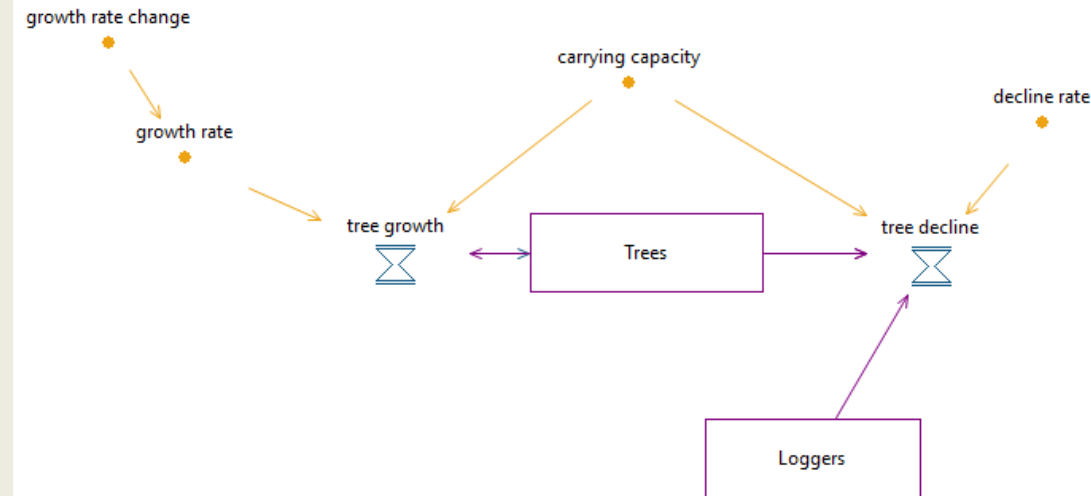
- Include a logistic growth function with the carrying capacity, which is the maximum size of the forest that the environment can sustain in the long run.
- Include a growth rate change variable using the TIME built-in

Case study assumption:

- Carrying capacity = 100 trees
- Growth rate change = 0.0002

New formulas

- $tree\ growth = Trees * growth\ rate * \left(1 - \frac{Trees}{carrying\ capacity}\right)$
- $tree\ decline = Trees * decline\ rate * \left(\frac{Trees}{carrying\ capacity}\right)$
- $growth\ rate = MAX(0, 0.05 - growth\ rate\ change * TIME)$



Run simulation: what can you see?

Exercise 5: Extending the model

- Loggers hiring depends on the number of trees available
- Loggers are more likely to get fired when there are not enough trees

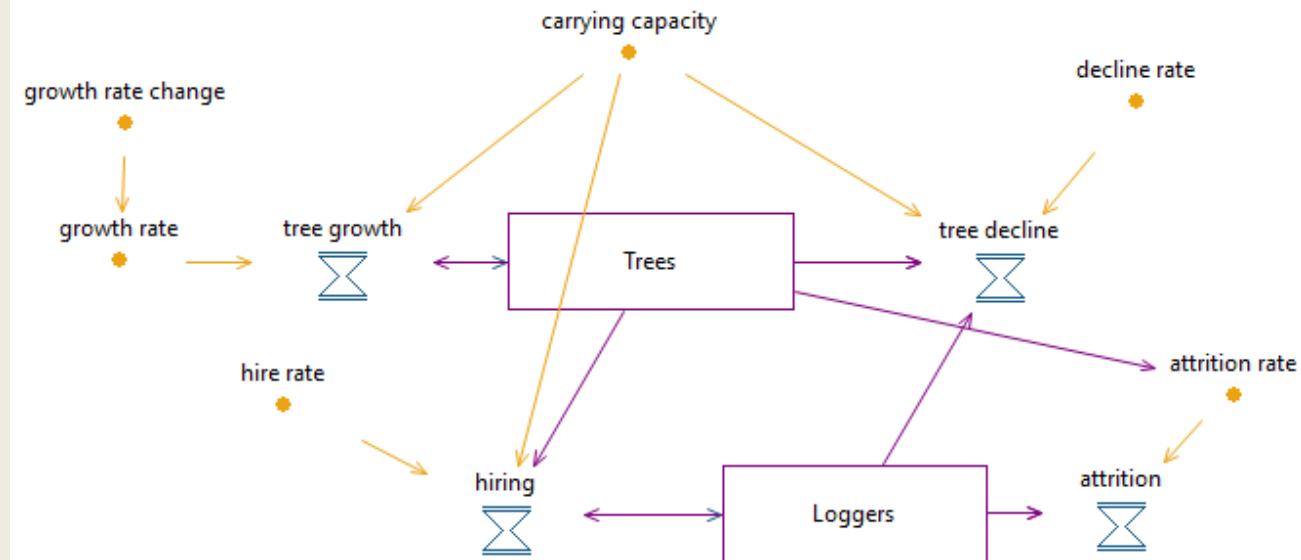
Case study assumption:

- Initial hire rate = 0.12
- Initial attrition rate = 0.07

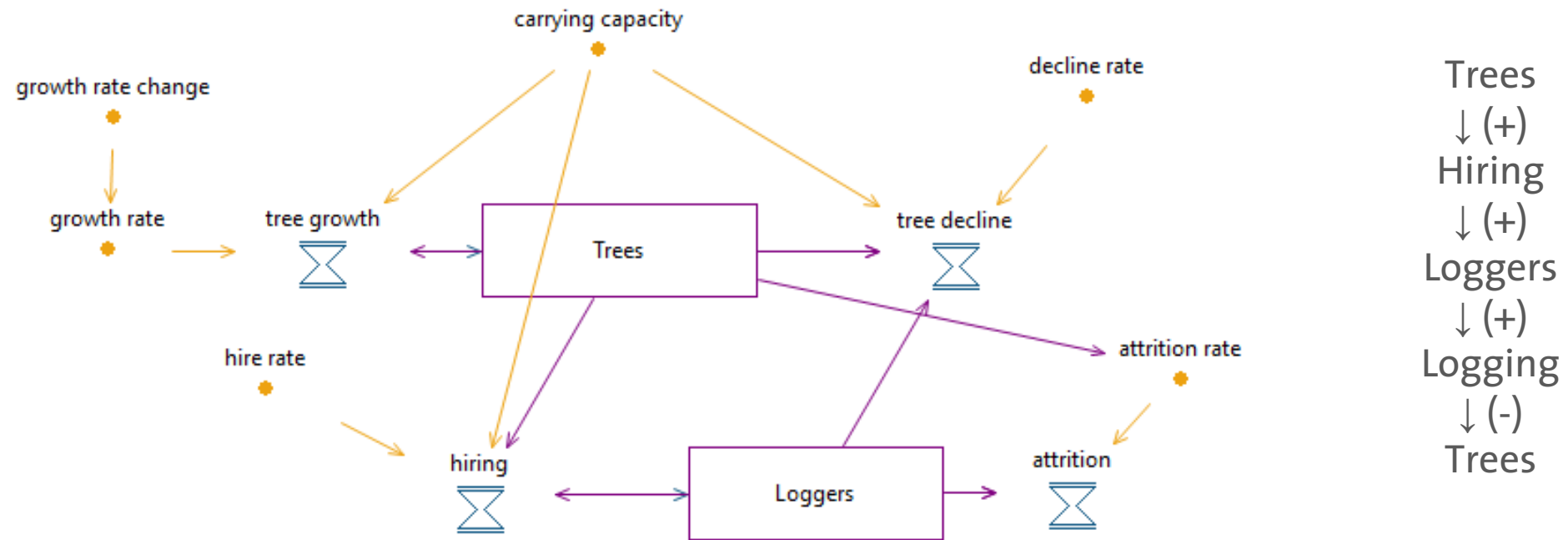
New formulas

- $\text{attrition rate} = \text{IF Trees} < 5 \text{ THEN } 0.8 \text{ ELSE } 0.07$
- $\text{hiring} = \text{Loggers} * \text{hire rate} * \left(\frac{\text{Trees}}{\text{carrying capacity}} \right)$
- $\text{attrition} = \text{Loggers} * \text{attrition rate}$

Run simulation: what can you see?

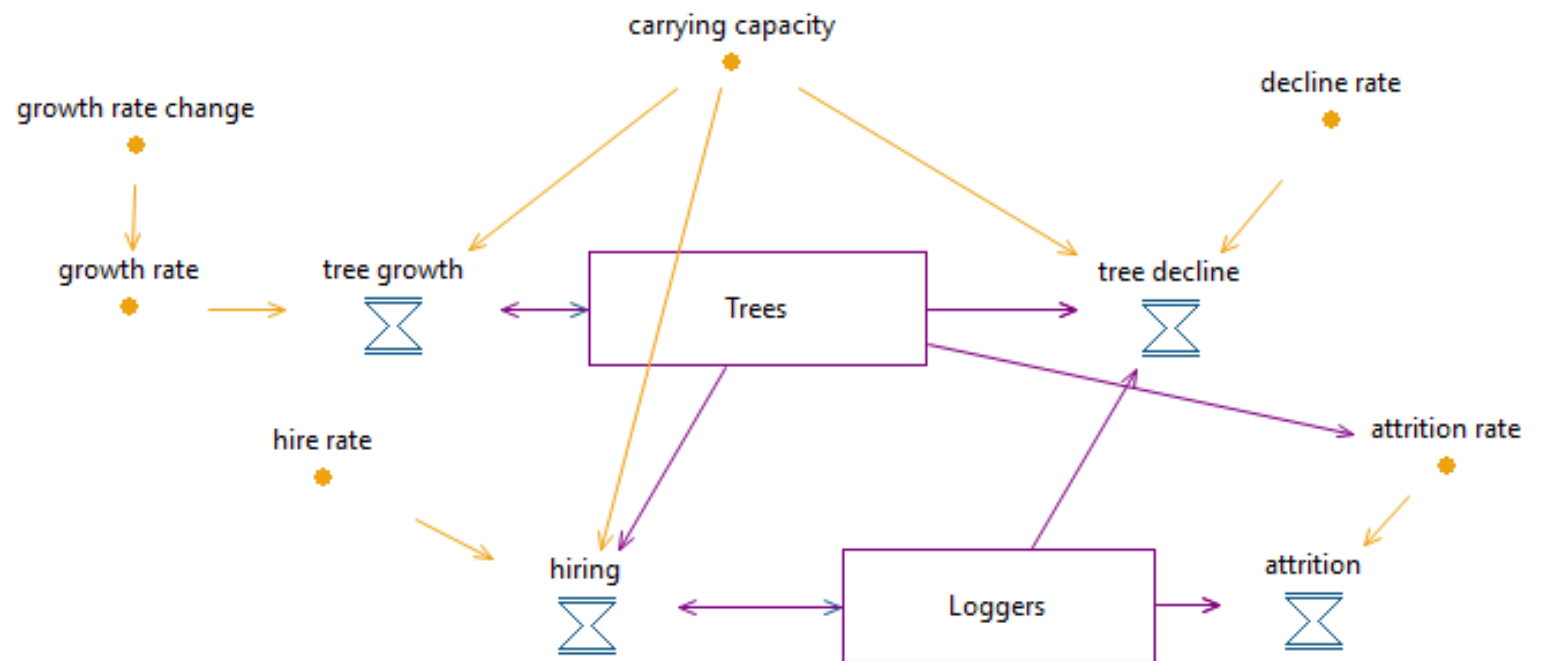


Feedback loop

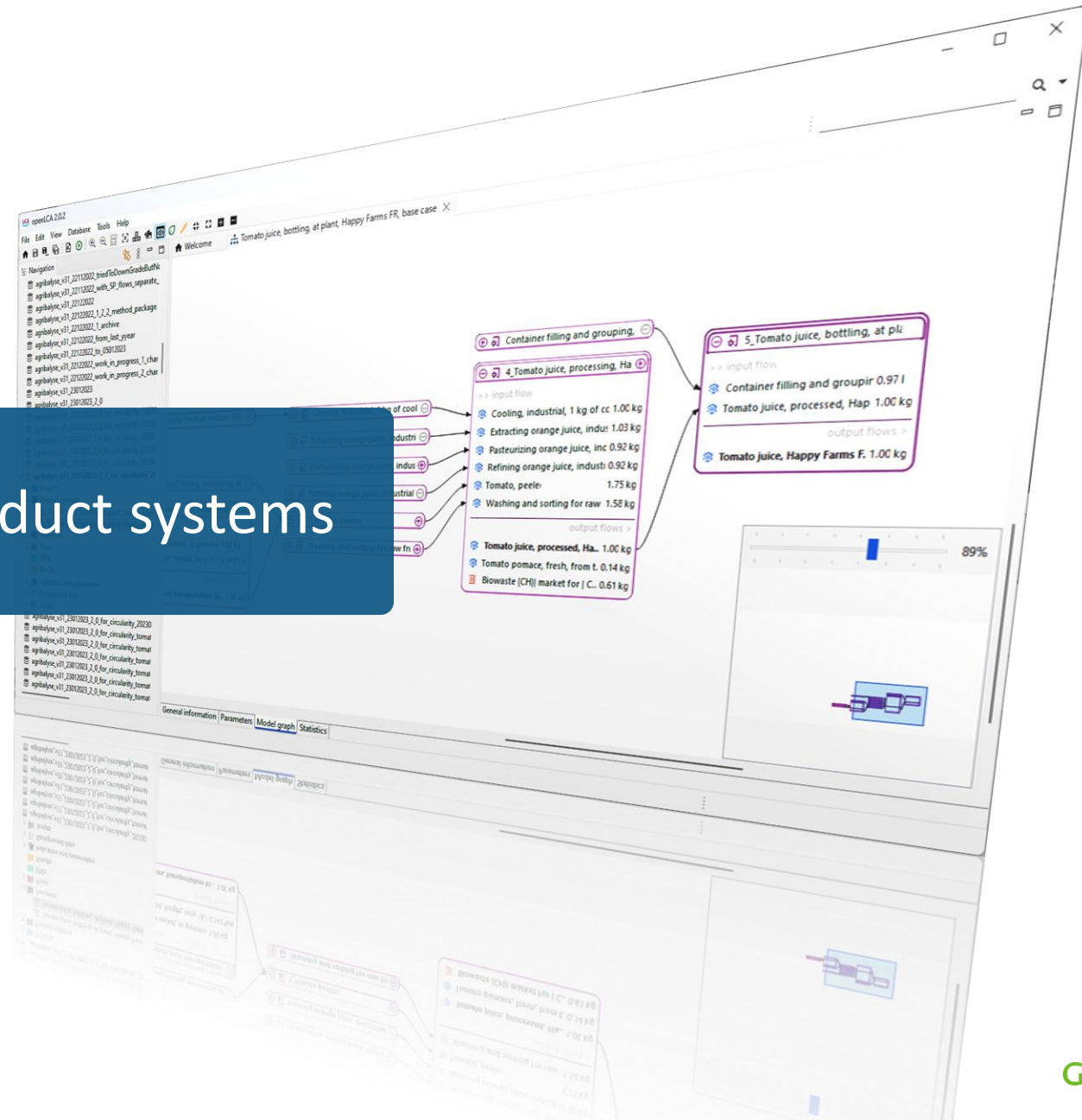


Options for exploring scenarios

- What happens when certain parameter values are changed? E.g. the decline rate
- Include delays or shocks
- Include thresholds



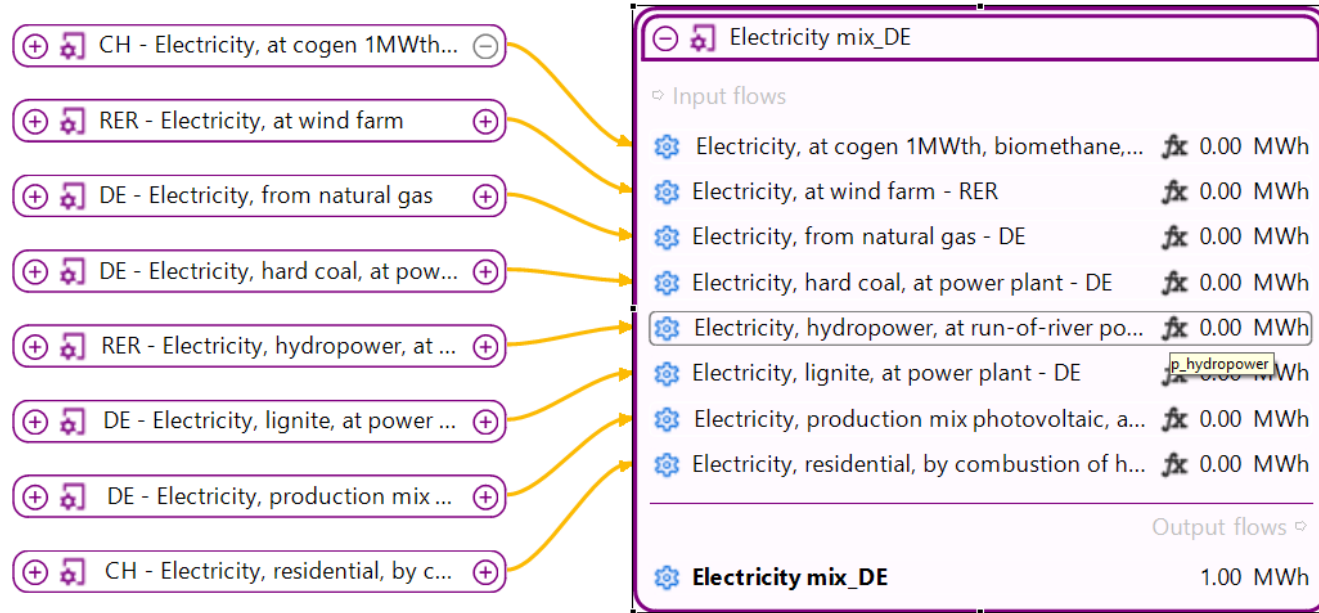
Connecting product systems



Connecting product systems

- Connecting a system dynamics model with an LCA product system is possible through linking parameters
- In each simulation step, it passes the calculated values of SD variables (stocks, rates, auxiliaries) into the LCA models through parameters which are then calculated with these parameter values.
- It will then produce a sequence of LCA results for the respective simulation steps.

Product systems with parameters

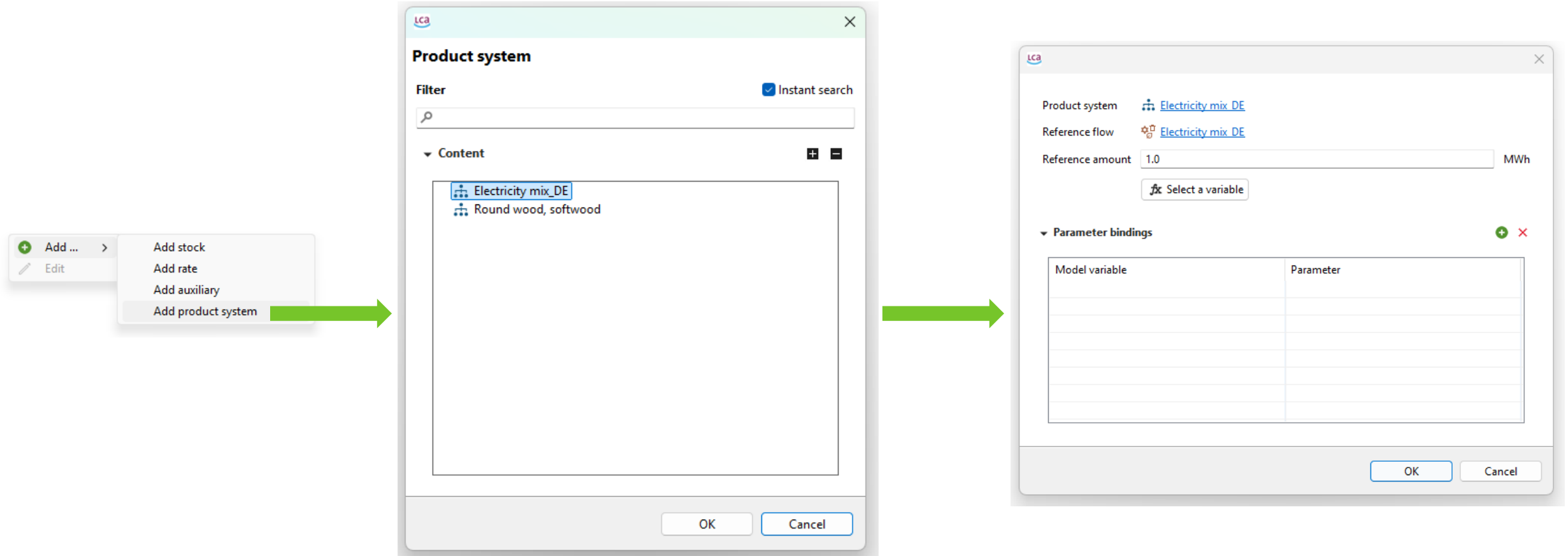


Inputs/Outputs - Electricity mix_DE

Inputs

Flow	Amount	Unit
Electricity, at cogen 1MWth, biomethane, a...	p_biomethane	MWh
Electricity, at wind farm - RER	p_wind	MWh
Electricity, from natural gas - DE	p_naturalgas	MWh
Electricity, hard coal, at power plant - DE	p_hardcoal	MWh
Electricity, hydropower, at run-of-river pow...	p_hydropower	MWh
Electricity, lignite, at power plant - DE	p_lignite	MWh
Electricity, production mix photovoltaic, at ...	p_solar	MWh
Electricity, residential, by combustion of hy...	p_hydrogen	MWh

Adding product systems



Adding product systems

The screenshot illustrates the process of adding product systems in the LCA software. The main window shows the 'Product system' set to 'Electricity mix DE', the 'Reference flow' set to 'Electricity mix DE', and the 'Reference amount' set to '1.0 MWh'. The 'Parameter bindings' section is currently empty. A green arrow points from the 'Add parameter binding' button to a secondary 'Variable binding' dialog. This dialog shows a list of 'Model variable' options, including 'electricity share biomethane', and a list of 'System parameter' options, including 'p_biomethane'. A green arrow points from 'electricity share biomethane' to 'p_biomethane', indicating the selection of this parameter for binding.

Product system [Electricity mix DE](#)

Reference flow [Electricity mix DE](#)

Reference amount 1.0 MWh

[Select a variable](#)

Parameter bindings

Model variable	Parameter

[Add parameter binding](#)

Variable binding

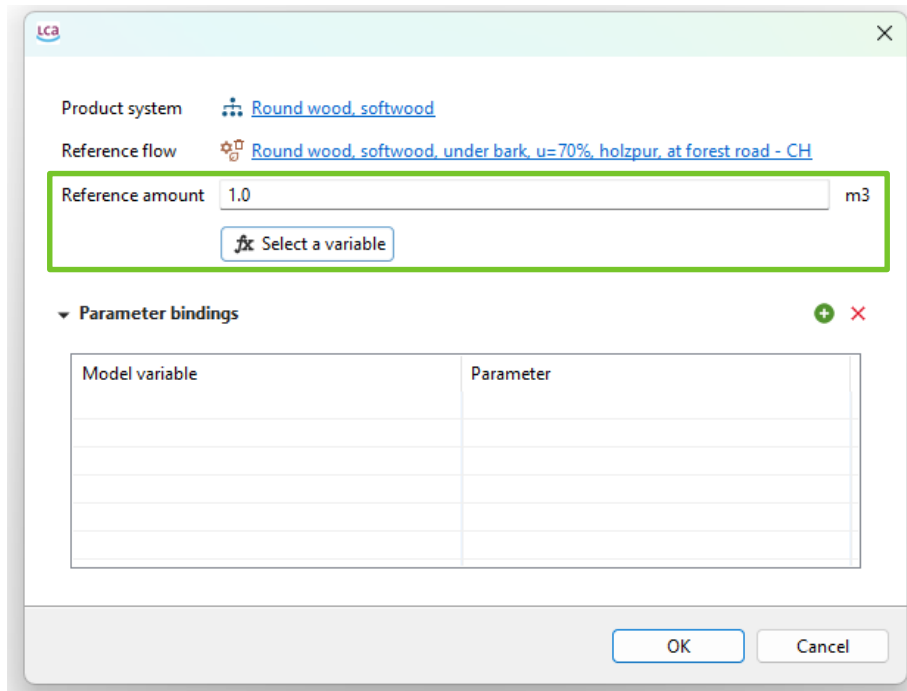
Model variable

- ☒ electricity share biomethane
- ☒ electricity share hard coal
- ☒ electricity share hydrogen
- ☒ electricity share hydropower
- ☒ electricity share lignite
- ☒ electricity share natural gas
- ☒ electricity share solar
- ☒ electricity share wind

System parameter

- ☒ compliant
- ☒ p_biomethane
- ☒ p_hardcoal
- ☒ p_hydrogen
- ☒ p_hydropower
- ☒ p_lignite
- ☒ p_naturalgas
- ☒ p_solar
- ☒ p_wind

Adding product systems

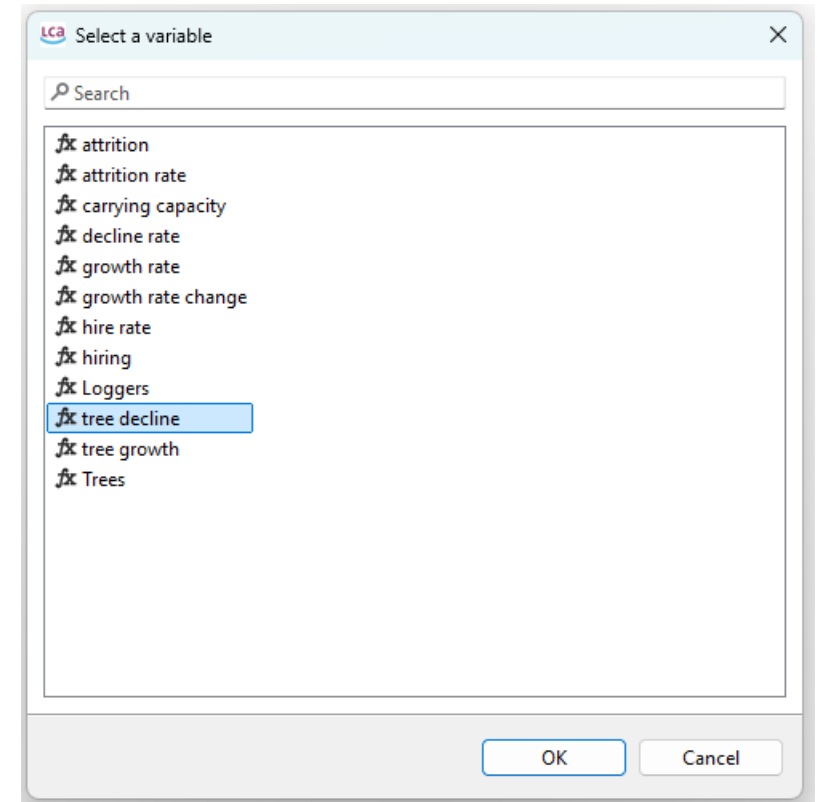


The 'LCA' dialog box is shown with the following configuration:

- Product system:** [Round wood, softwood](#)
- Reference flow:** [Round wood, softwood, under bark, u=70%, holzpur, at forest road - CH](#)
- Reference amount:** 1.0 m3
- Select a variable:** A button with a tree icon and the text 'Select a variable'.
- Parameter bindings:** A section with a table for defining model variables and their parameters.

Model variable	Parameter

Buttons: OK, Cancel



The 'Select a variable' dialog box displays a list of variables for selection:

- attrition
- attrition rate
- carrying capacity
- decline rate
- growth rate
- growth rate change
- hire rate
- hiring
- Loggers
- tree decline** (highlighted)
- tree growth
- Trees

Buttons: OK, Cancel

Selecting an LCIA method

Welcome *Forest model

System dynamics model: Forest model

▼ General information

Name: Forest model

Start time: 0.0 years

Stop time: 100.0 years

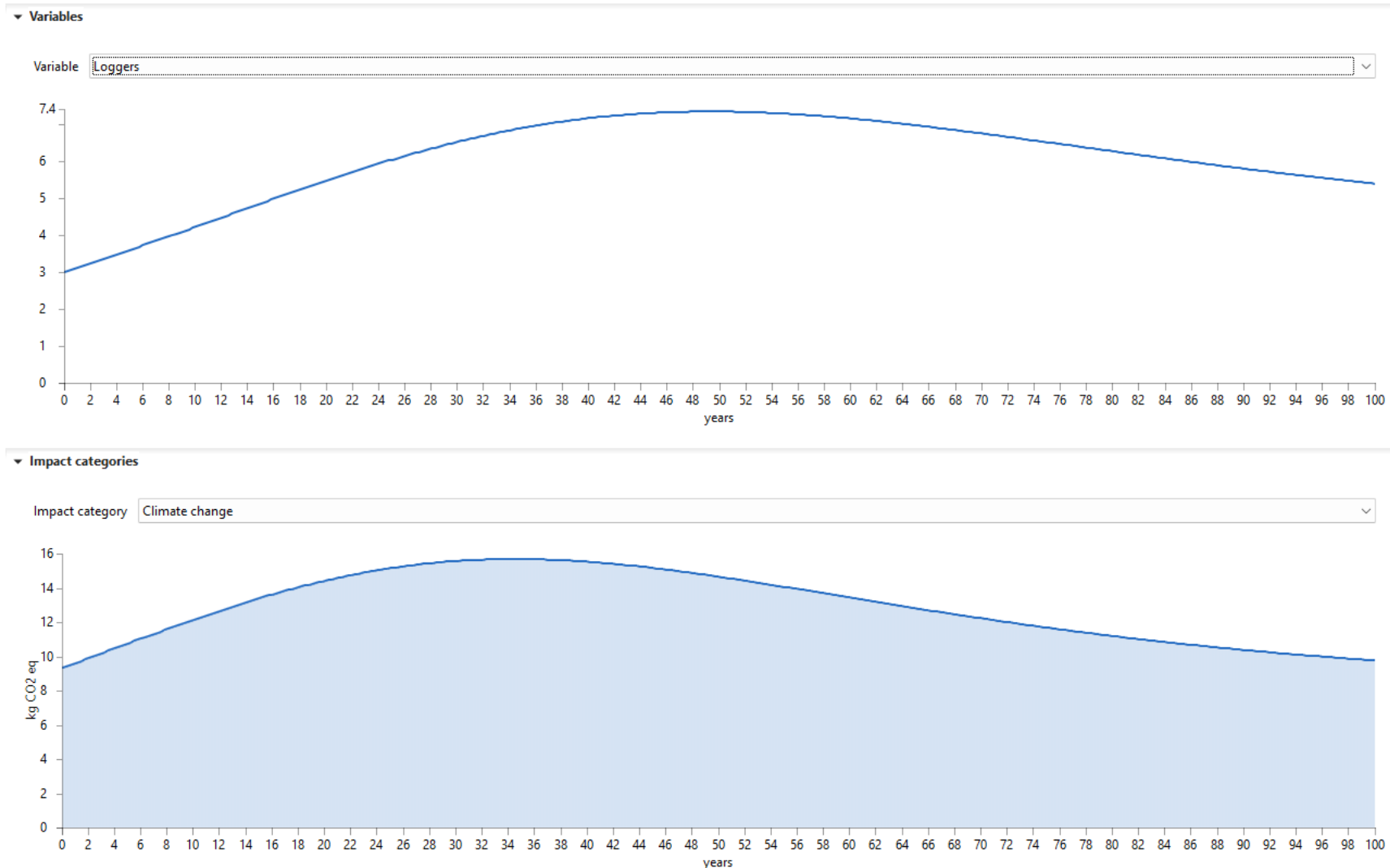
Δt : 0.25 years

Time unit: years

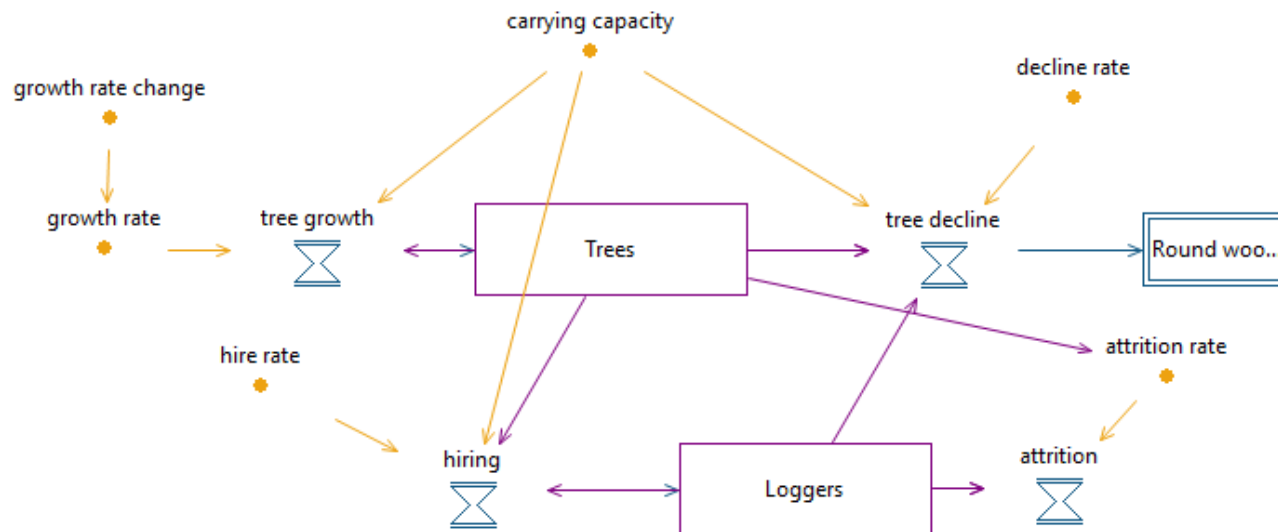
Impact assessment method: EF 3.1 Method (adapted)

EF 3.1 Method (adapted)	openLCA LCIA Methods 2.8.0 adapted
Environmental Prices	openLCA LCIA Methods 2.8.0 adapted
EPD 2018	openLCA LCIA Methods 2.8.0 adapted
EPS 2015d	openLCA LCIA Methods 2.8.0 adapted
EPS 2015dx	openLCA LCIA Methods 2.8.0 adapted
Hoekstra et al 2012 (Water Scarcity)	openLCA LCIA Methods 2.8.0 adapted
ILCD 2011 Midpoint+	openLCA LCIA Methods 2.8.0 adapted

Simulation results



Exercise 6: Connecting a product system



Case study assumption:

- Connect the product system “Round wood, softwood”
- Link it to the model by making the reference amount “tree decline”
- Select the method “EF 3.1 Method”

Run simulation: what can you see?

Name	Forest model
Start time	0
Stop time	100
Time step (Δt)	1.0
Time unit	years

Exporting results



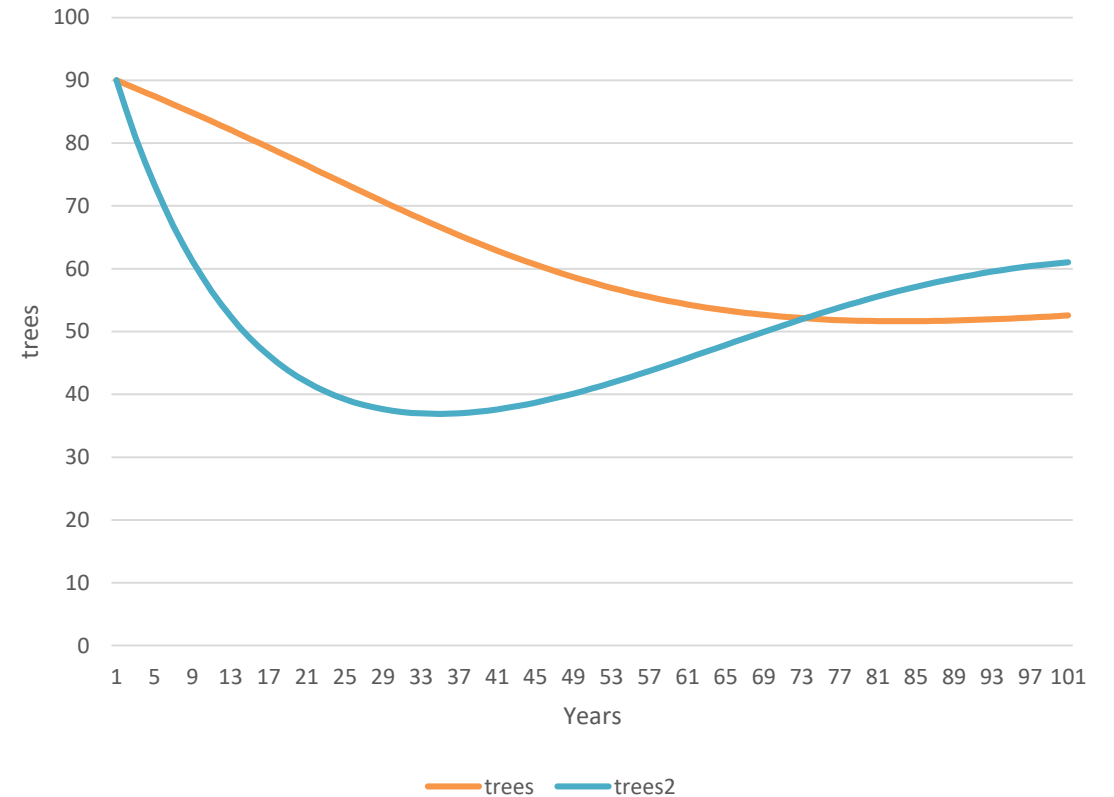
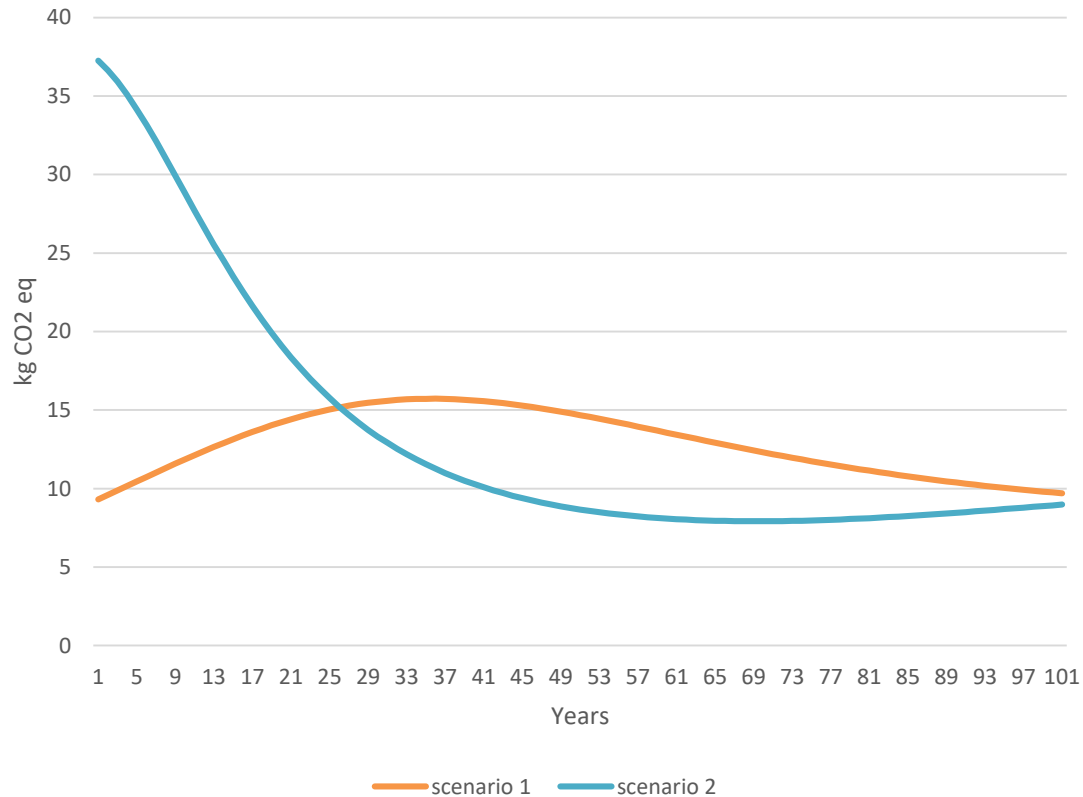
Exporting results

The screenshot shows a web application interface with three tabs: 'Welcome', 'Forest model', and 'Forest model - Simulation result'. The 'Forest model - Simulation result' tab is active, displaying the title 'Simulation results: Forest model'. Below this, there is a section titled 'General information' with a dropdown arrow. This section contains four input fields: 'Model' (Forest model), 'Variables' (Stocks: 2, Rates: 4, Aux: 6), 'Impact categories' (25), and 'Iterations' (101). At the bottom of this section, there is a button labeled 'Export results...' with a small icon of a document with an 'x' on it. The button is highlighted with a green rectangular border.

Field	Value
Model	Forest model
Variables	Stocks: 2, Rates: 4, Aux: 6
Impact categories	25
Iterations	101

 Export results...

Comparing scenarios



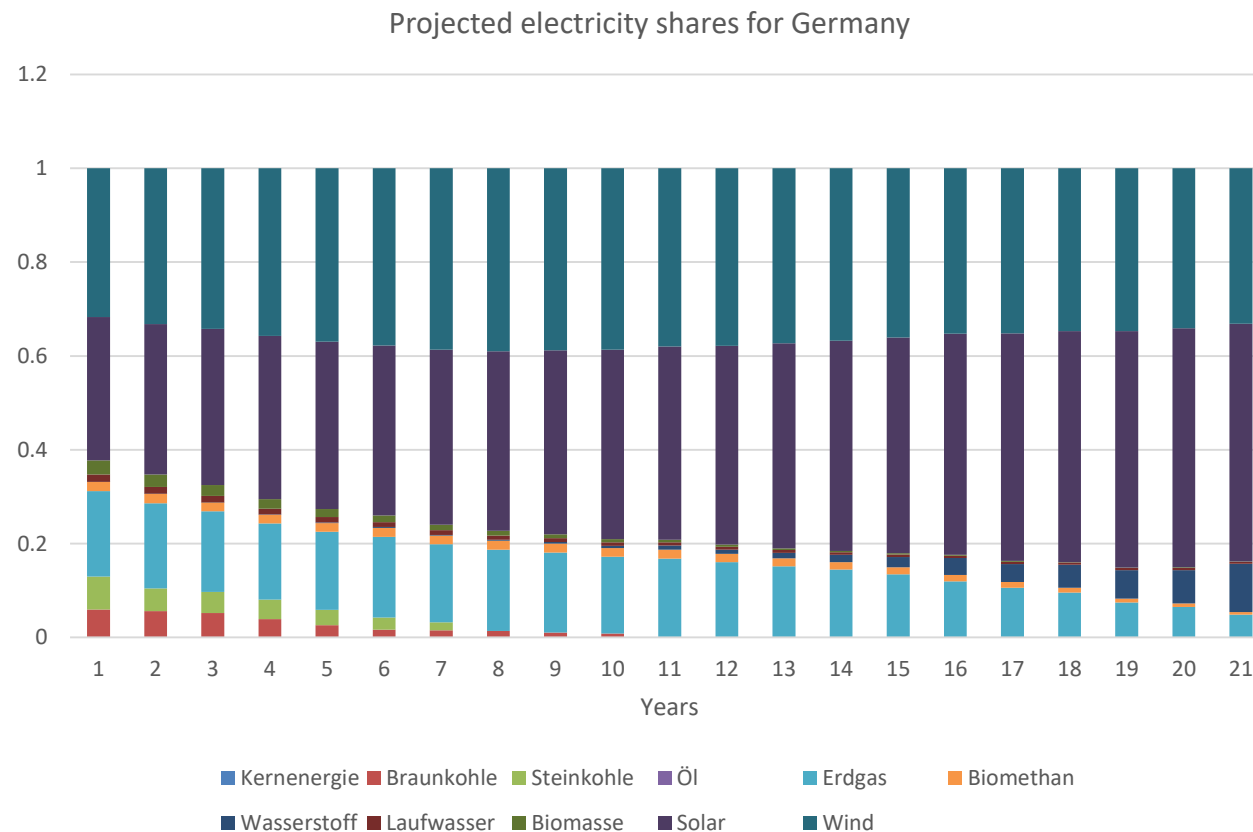
- Scenario 1: decline rate = 0.5
- Scenario 2: decline rate = 2

Other use cases



System dynamics in openLCA

- Simple use of the Lookup option: exploring electricity mix variations



LCA

Name: electricity share hydrogen

Unit:

Non-negative: ☐

Type: Lookup function

Equation for x: TIME

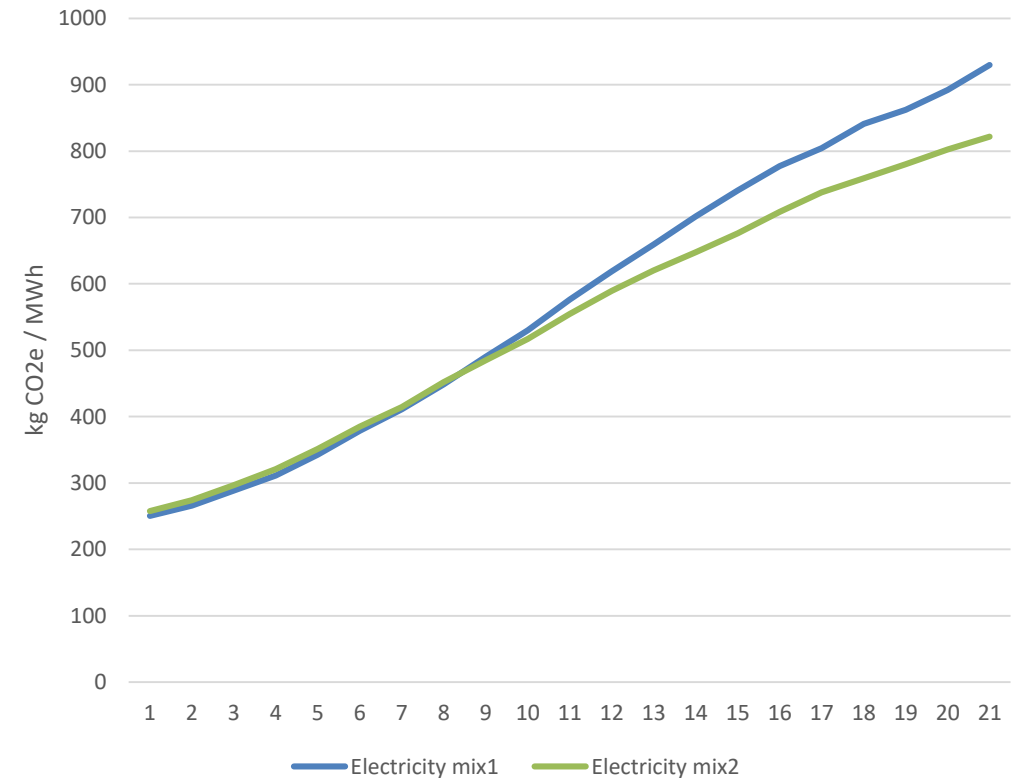
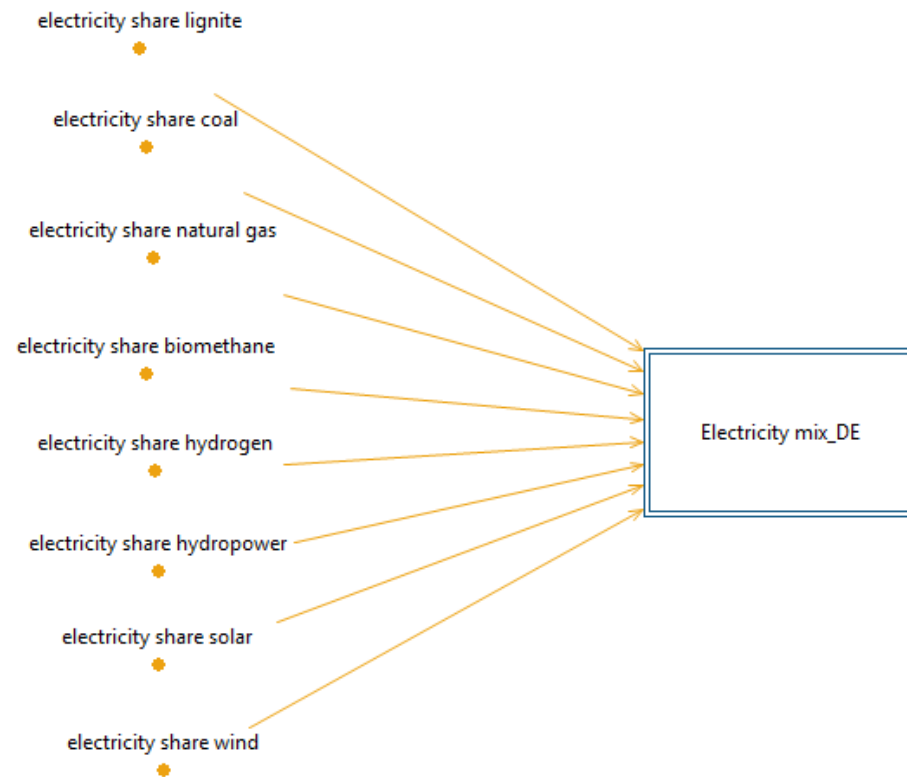
Lookup values - linear interpolation

x	y
0.0	1.59699E-5
1.0	1.50335E-5
2.0	1.38713E-5
3.0	6.74712E-4
4.0	0.001448454
5.0	0.002012317
6.0	0.001854628
7.0	0.001980273
8.0	0.003385528
9.0	0.004693117
10.0	0.00901465
11.0	0.00841493
12.0	0.011407046

OK Cancel

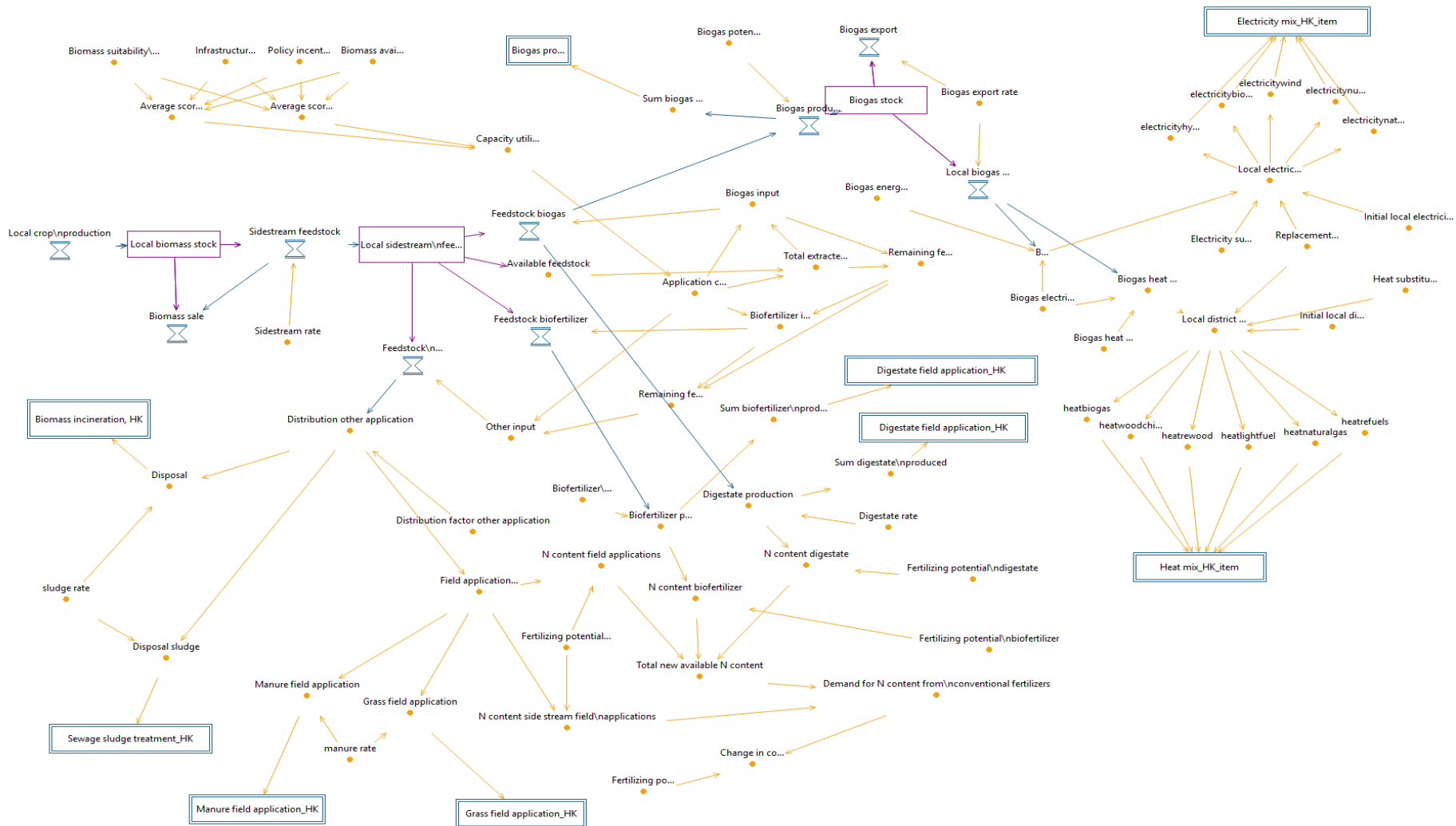
System dynamics in openLCA

- Simple use of the Lookup option: exploring electricity mix variations



System dynamics in openLCA

- Regional impacts of introducing biogas capacities

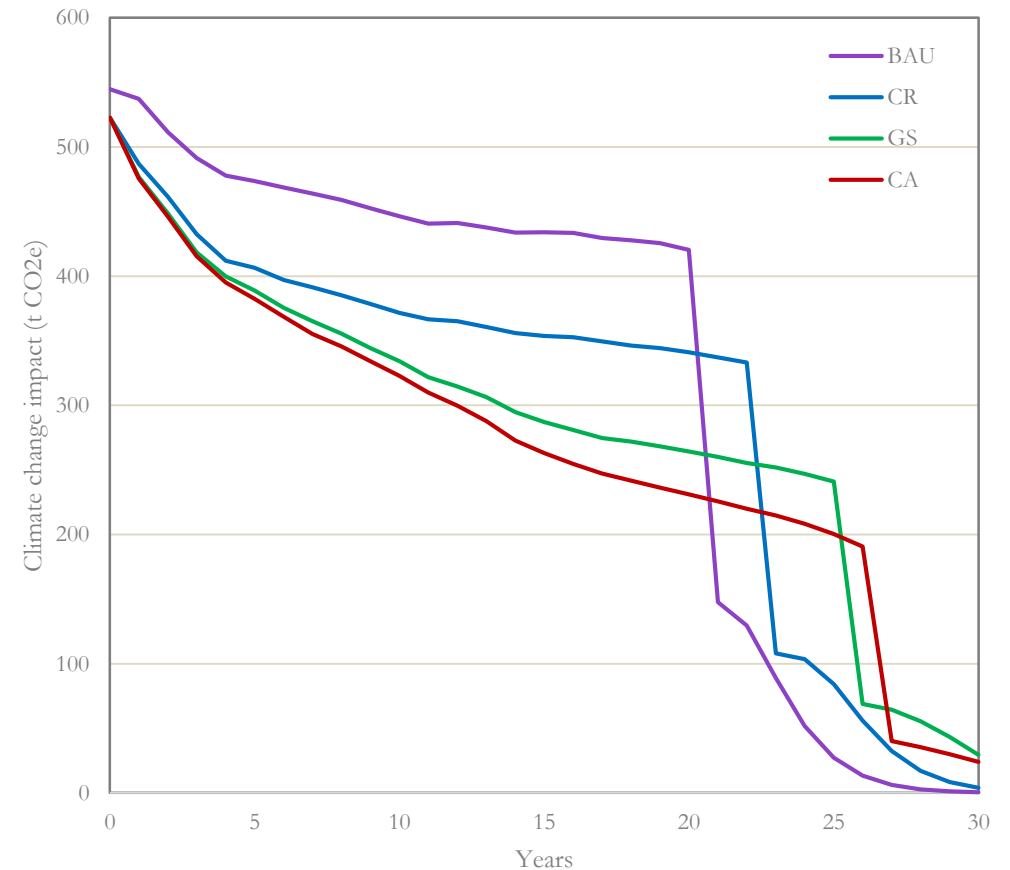
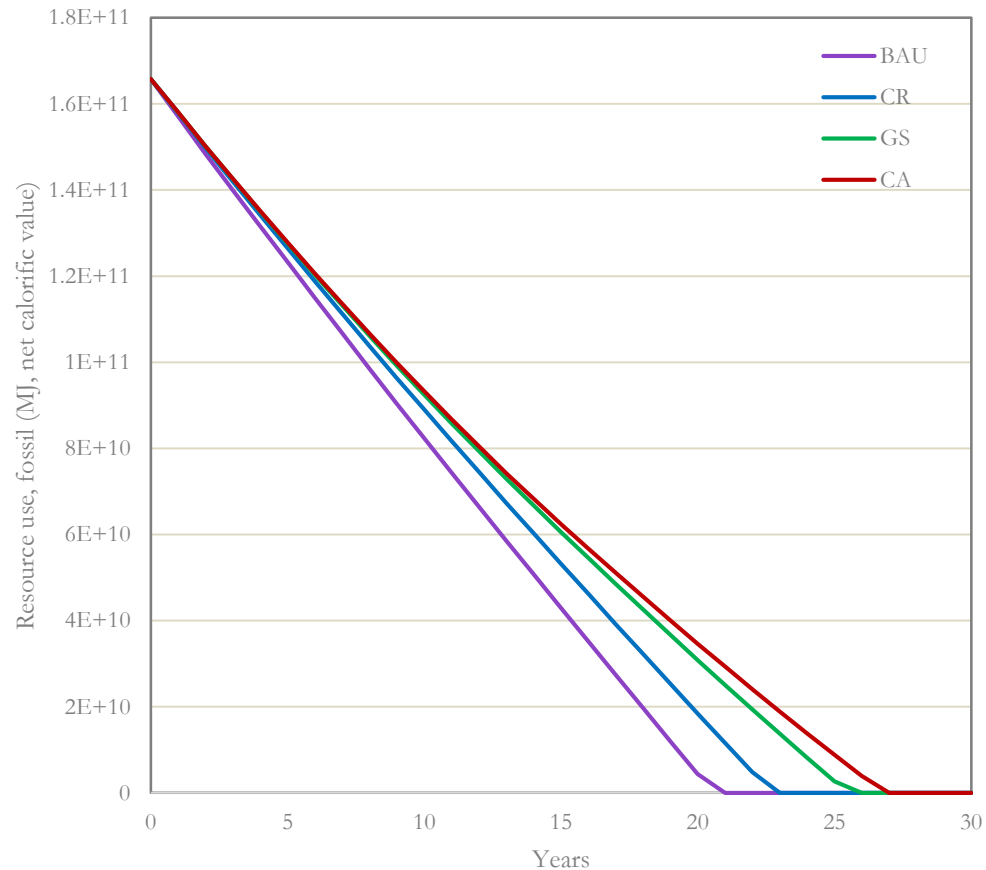


- Modelling market plastic substitution



System dynamics in openLCA

- Including planetary boundaries as stocks



Other use cases?

Reflections

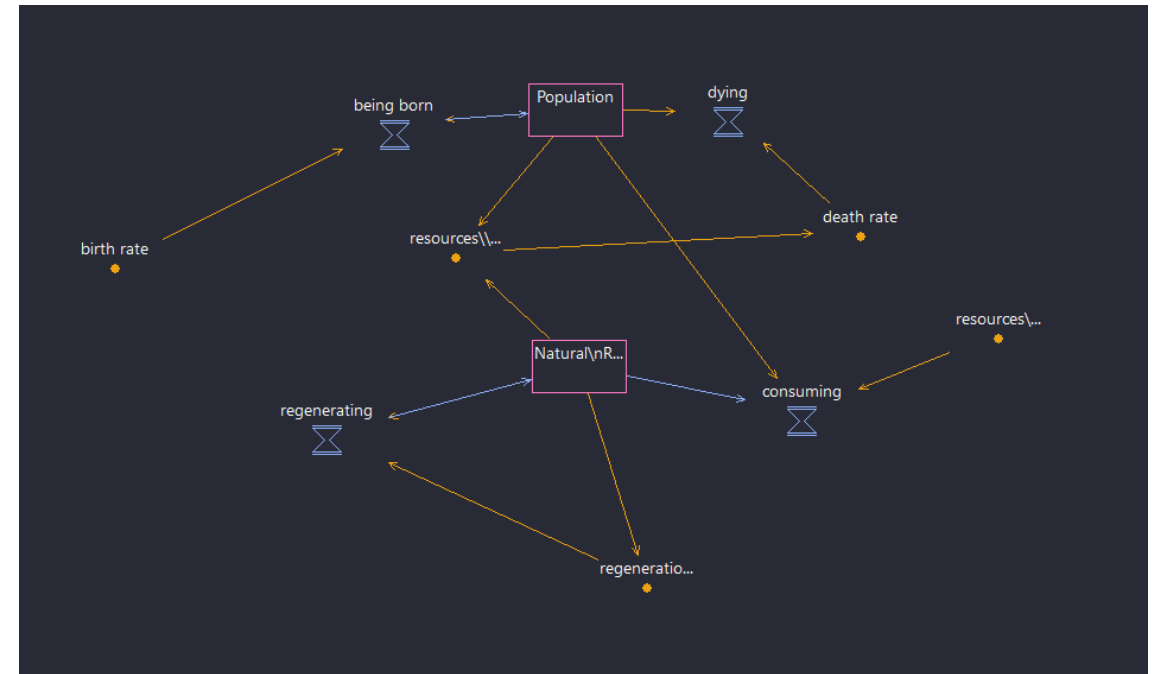
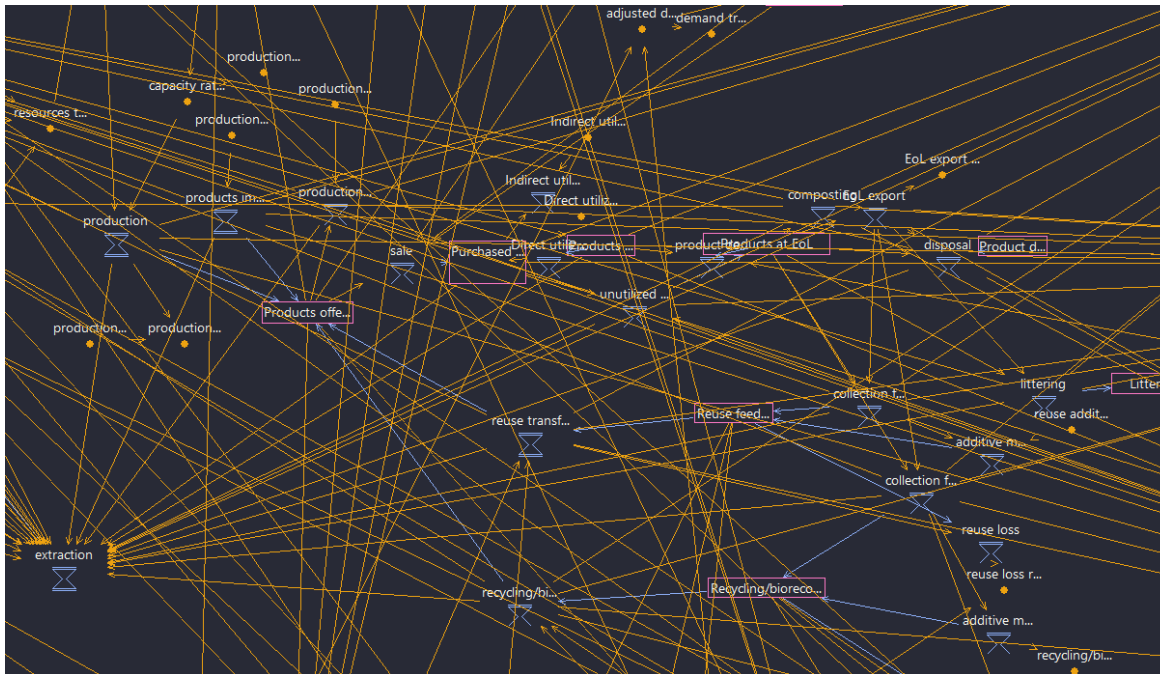


Potential

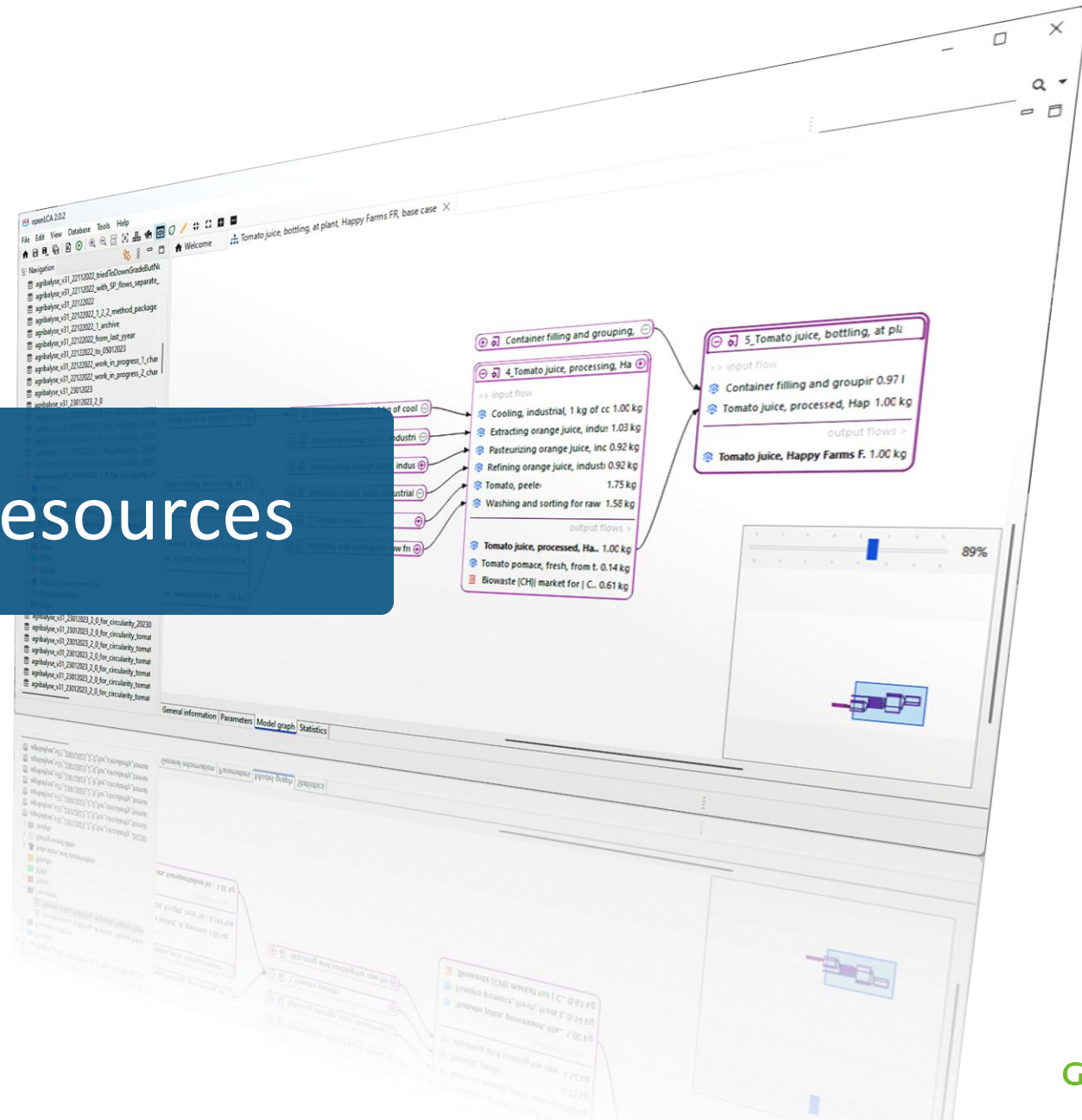
- Flexible dynamic modelling (of any system)
- Flexible parameterization of LCA models to allow all types of time-dependent impact calculations
- User-defined future transition scenarios – export results and comparing scenarios
- Dynamic modelling with socio-economic drivers
- Feedback loops
- Including market effects and indirect consequences (allows considering rebound effects)
- Resource constraints and planetary boundaries – modelling with stocks

Limitations

- Big models require a lot of data
- SD modelling often done by experts
- Still some limitations and issues with the current openLCA implementation (but we are working on it) – also need more people to use and test the feature



openLCA Resources



openLCA manual

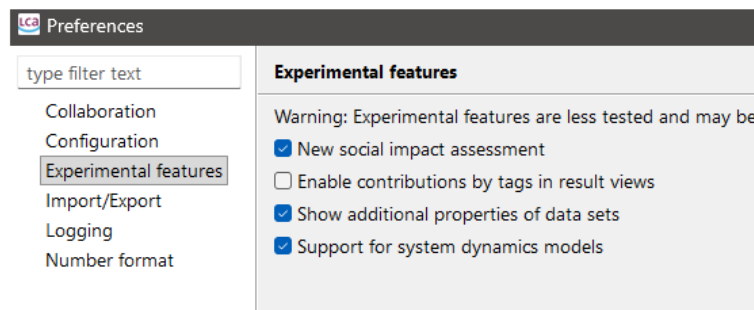
5. Running openLCA for the first time	>
6. Databases	>
7. Flows	>
8. Processes	>
9. Product Systems	>
10. LCIA methods and categories	>
11. Calculation and Result Analysis	>
12. Projects	>
13. Waste modelling	>
14. Allocation	>
15. Parameters	>
16. Background data	>
Guidance for LCA practitioners	
17. Model validation in openLCA	>
18. Data aggregation in openLCA	>
The advanced knowledge base	
19. Advanced Topics	>
19.1. Regionalized LCA	>
19.2. Life Cycle Costing	>
19.3. Social aspects	>
19.4. Data Quality	>
19.5. Monte Carlo simulation	>
19.6. Modelling with time parameter	>
19.7. System dynamics	>
20. Environmental Product Declarations (EPDs)	>
21. Libraries	>
22. Scripting in openLCA	>



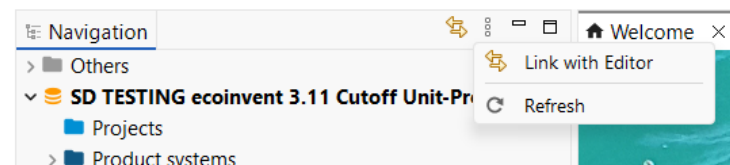
Current implementation of system dynamics in openLCA

System dynamics allows to model non-linear behaviour of complex systems over time using stocks, flows, and feedback loops. It makes it possible to model product systems in broader system surroundings and further define dynamic relationships of system variables. In openLCA you can now link your LCA models to system dynamics models, making it possible to calculate dynamic impact results over time.

First, you need to enable "Support for system dynamics models" in the preferences (do not forget to press "Apply" or "Apply & Close" in that dialog):



Then, refresh the navigation via F5 or the refresh button:



You should see a folder "System dynamics models" in your database folders now:

openLCA nexus

openLCA Nexus

Databases

Services

Models

Utilities

LCA data search

Map

Documents

FAQs

About

Register

Login

0



openLCA Nexus

Your source for LCA and sustainability data.

Models

- ecoinvent models
- Social models
- System dynamics models
- Other free models

Models are listed by the database they use. Each model contains one or several openLCA product systems, i.e. life cycle models, and a report and documentation



ecoinvent models

Free and for purchase models using the ecoinvent database. Please note that you need an ecoinvent licence to download this model.

Browse



Social models

All the documentation for these models is available for free under the "Documents" tab.

Browse



System dynamics models

Free system dynamics models using free databases.

new

Browse

GreenDelta

sustainability consulting + software



Thank you!

Alexander Koch, koch@greendelta.com

GreenDelta GmbH

Alt-Moabit 130/131, 10557 Berlin

www.greendelta.com

Picture: Julia Cilleruego Palomero